

# RELAYSPEC: RETARGETING FROZEN SPECULATIVE DRAFTERS WITH SMALL-DATA LINEAR INTERFACES

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## ABSTRACT

Feature-conditioned speculative drafters depend on the hidden states of the model used to train them. Reusing such a drafter with a larger target can therefore require running its source model alongside that target. RelaySpec replaces this dependency with a learned linear interface while keeping the target and drafter frozen. Across DFlash and EAGLE-3 drafters retargeted from Qwen3-4B to 8B and 14B, the original 4,096-record fits achieve 2.35–5.11 times autoregressive throughput on MATH-500. Separate EAGLE-3 measurements reduce peak allocated memory by 7.63–7.80 GiB relative to source reuse. Controlled scaling studies show that the interface can be learned from substantially fewer records: frozen 512-example maps retain 97.5% (DFlash) and 98.9% (EAGLE-3) of their 2,048-example maps’ throughput on 256 held-out GSM8K questions with 8B targets. Both confidence intervals exceed 95% retention, although tight accuracy noninferiority remains unresolved. The simple dense map has the highest mapper throughput in the completed 8B long-output capacity comparisons. Wider MLPs and longer fitting do not consistently improve decoding. Under matched warm-training budgets, feature regression outperforms the tested connector-CE and LoRA alternatives. These results identify a low-cost reuse regime and its limits, including task-dependent slowdowns relative to source reuse, optimization sensitivity and finite-precision output differences.

## 1 INTRODUCTION

A developer may have a DFlash drafter trained for Qwen3-4B but want to serve Qwen3-8B or Qwen3-14B. The new target can check its proposals, but supplies different hidden states. This creates a deployment choice: train a new target-specific drafter, or reuse the existing drafter while paying for its source model to provide compatible features. RelaySpec addresses deployments that seek to reuse this training investment with limited additional fitting work. It learns a replacement for the source conditioning path, allowing the new target to supply the drafter’s input. The practical goal is faster generation than plain AR while retaining much of a native target-specific drafter’s throughput.

Speculative decoding accelerates autoregressive (AR) generation by checking several proposed tokens in one target pass (Leviathan et al., 2023). DFlash and EAGLE-3 improve proposals using features computed inside a language model (Chen et al., 2026; Li et al., 2025). We call the model that provided these features during drafter training the *source*. The *target* is the model whose token choices govern deployment. Running the source alongside a new target supplies compatible features, but also retains its transformer computation and memory.

RelaySpec runs the source and new target on shared text during fitting. It learns a linear map from the target’s hidden states to the context that the source supplies to the drafter. At inference, the target already computes these states when checking proposals. Applying the map supplies the next drafting input without another source-transformer pass. The existing target verifier still decides which tokens to commit.

Only the map is trained. The target, drafter, inherited embedding and vocabulary projection stay fixed. The practical value is retaining an existing drafter’s training investment when upgrading the verifier, while eliminating the source transformer from each generation cycle. A small interface is useful only

if it preserves enough proposal quality to make that removal worthwhile. Our experiments test this directly.

Our contribution has three parts. First, we identify and replace the source-conditioning boundary in two released drafter families, obtaining 2.35–5.11 times AR throughput on MATH-500 and separately saving 7.63–7.80 GiB of allocated memory. Second, frozen-checkpoint confirmation shows that 512 fitting examples retain 97.5% and 98.9% of the 2,048-example maps’ speed. Third, we test whether simplicity sacrifices performance: matched linear/MLP capacity sweeps, longer fits, regularization and measured-budget CE/LoRA comparisons make the dense linear interface a strong empirical choice. It achieves the highest mapper throughput in the completed 8B long-output capacity comparisons, and feature regression wins every tested adaptation budget. The insight is that a frozen drafter’s useful prediction behavior can survive a cheap change of conditioning interface. Increasing the expressiveness or training work of that interface need not improve decoding. Layer ablations identify a compact late-layer interface, and native-interface compression shows that this analysis also applies without retargeting. Cross-family, domain-composition and 14B experiments identify the limits of these findings.

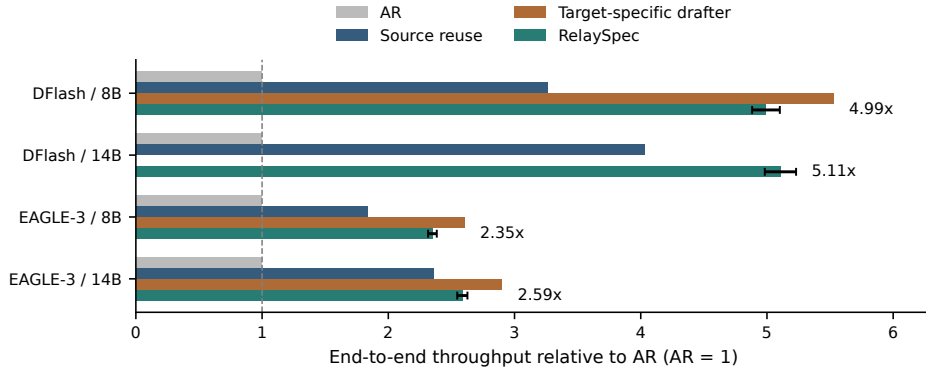


Figure 1: MATH-500 throughput relative to plain AR, measured within each paired run. RelaySpec reuses the 4B drafter. The native control uses a drafter released for the evaluated target. Whiskers show RelaySpec’s 95% paired intervals. Native controls show the three evaluated releases. Each runtime uses one request per GPU.

## 2 RELATED WORK AND POSITIONING

**Retargeting an inherited drafter.** DFlash and EAGLE-3 condition their proposals on target features (Chen et al., 2026; Li et al., 2025), tying an inherited drafter to its training interface. TriSpec already supports adapter-only finetuning into a fixed EAGLE-family drafter, mapping proxy features to the expected feature space (Jiang et al., 2026). Adapter-only tuning is therefore not our novelty claim. TriSpec uses a proxy to approve some tokens without full-target verification. RelaySpec instead removes the source transformer from generation, retargets its inherited drafter to a larger verifier, and retains target-controlled acceptance. Our contribution is this deployment setting and its measured data, capacity and source-removal behavior across two drafting families.

**Target-independent drafting and dynamic steering.** PARD adapts an autoregressive draft model for parallel token prediction and reuses it across a target family (An et al., 2026). Thus reuse across targets is established prior work. Its target-independent drafter offers zero additional target-specific fitting at deployment. RelaySpec instead makes a small target-specific investment to reuse a feature-conditioned checkpoint and exploit the new target’s internal states.  $SD^2$  is a closer alignment alternative: it computes steering from verifier states and injects it into a pretrained drafter (Berdoz et al., 2026). Our fitted input map replaces source conditioning, whereas  $SD^2$  learns steering within the drafter. The experiments include its public frozen-drafter configuration and PARD’s released checkpoint, with their own matched AR controls (Appendix D). These comparisons expose practical

alternatives without attributing differences in runtime or inherited models to the alignment mechanism alone.

**Distillation and drafter adaptation.** OmniDraft uses online hybrid distillation and an n-gram cache to address cross-vocabulary deployment (Ramakrishnan et al., 2025). RelaySpec studies offline fitting with a shared vocabulary, so it does not replace heterogeneous-vocabulary verification (Timor et al., 2025a). RepSpec changes draft training through structural re-parameterization (Huo et al., 2026). VSD optimizes sequence acceptance, and the Draft-OPD preprint studies on-policy teacher feedback (Zou et al., 2026; Lei et al., 2026). These approaches improve a drafter’s learned prediction behavior. RelaySpec holds that behavior fixed and regresses its input context. Our connector-CE and rank-32 LoRA experiments test whether updating token prediction is a better use of the same measured warm-training budget. Feature fitting wins every tested budget comparison (Table 4), which supports this inexpensive adaptation recipe without ruling out other distillation objectives or longer full-drafter training.

**Frozen interfaces and the meaning of feature alignment.** Adapters, low-rank updates and model stitching already adapt or connect frozen networks (Houlsby et al., 2019; Hu et al., 2022; Bansal et al., 2021). A linear connector alone is therefore not the novelty. The contribution is locating an existing speculative decoder’s source-conditioning boundary, learning its replacement while preserving the target and drafter, and establishing the data, capacity and deployment consequences. Successful stitching need not imply equivalent information (Smith et al., 2025). We consequently test decoded speed and quality, equal-capacity linear and nonlinear maps, saved fitting trajectories and frozen-checkpoint confirmation. The observed gaps between training error, validation error and decoding speed are central findings rather than evidence that representation similarity is sufficient.

**Proposal structure and execution are separate axes.** Medusa changes prediction heads, EAGLE-2 and OPT-Tree allocate draft trees, and SpecInfer organizes tree verification (Cai et al., 2024; Li et al., 2024b; Wang et al., 2025a; Miao et al., 2024). Self-speculative methods reuse parts of the target itself (Zhang et al., 2024; Elhoushi et al., 2024), while speculative speculative decoding overlaps drafting and verification (Kumar et al., 2026). RelaySpec changes who supplies the drafter’s input, not these execution policies. Its measured advantage is amortizing an inherited drafter with little fitting data and lower source-conditioning cost. We establish this on two drafter families and two target sizes, then identify when extra mapper capacity, optimization or broader training data helps and when it does not.

### 3 PRELIMINARIES

#### 3.1 DRAFTING AND TARGET VERIFICATION

An autoregressive model generates one token at a time. Blockwise and speculative methods propose several tokens, then check them in a target pass (Stern et al., 2018; Xia et al., 2023; Leviathan et al., 2023). In greedy decoding, the verifier commits the longest matching prefix and a target-selected next token. RelaySpec preserves this decoder and changes the features supplied to its drafter.

#### 3.2 THROUGHPUT AND ACCEPTED PROGRESS

Our primary metric is end-to-end output throughput: total generated tokens divided by total request time, including prompt processing. Let  $T_A$ ,  $T_R$  and  $T_N$  denote this throughput for plain AR, RelaySpec and the native target-specific drafter. We report

$$\text{AR speedup} = T_R/T_A, \quad \text{native throughput retained} = 100 T_R/T_N \%. \quad (1)$$

Throughput includes prompt processing. We also report summed AR time relative to summed relay time because output lengths can differ. Accepted progress counts tokens advanced per verification cycle, including the target-supplied token. Removing source computation saves cycle time but can reduce accepted progress. Appendix E specifies this tradeoff and the position-wise acceptance convention.

## 4 METHOD

### 4.1 FIT THE INPUT CONSUMED BY THE FROZEN DRAFTER

We run the source and new target on the same text and collect hidden states after five selected transformer blocks. Concatenating these states gives the target vector  $h_t$  and source vector  $s_t$  at position  $t$ . The drafter’s frozen projection  $F$  converts source features to its input width. We learn a bias-free matrix  $W$  that supplies this input from  $h_t$ . Figure 2 separates fitting from generation.

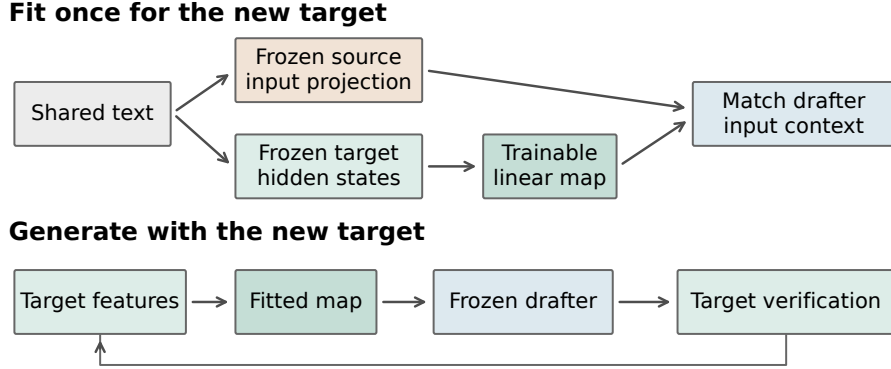


Figure 2: Fit once, then reuse the frozen drafter. The source and target process identical text during fitting. Only the map is updated. During generation, target features from committed positions feed the next draft.

DFlash consumes normalized projected features, while EAGLE-3 consumes the raw projection. Let  $N_{\text{in}}$  be fixed RMS normalization (Zhang & Sennrich, 2019) and  $N_{\text{out}}$  the released output normalization. Teacher and mapped contexts are

$$\begin{aligned} \text{DFlash: } c_t &= N_{\text{out}}(Fs_t), & \hat{c}_t &= N_{\text{out}}(WN_{\text{in}}(h_t)), \\ \text{EAGLE-3: } c_t &= Fs_t, & \hat{c}_t &= Wh_t. \end{aligned} \quad (2)$$

We fit  $W$  by minimizing the squared context error relative to the teacher’s squared magnitude. For a record with  $n$  token positions,

$$\mathcal{L}(W) = \frac{1}{n} \sum_{t=1}^n \frac{\|\hat{c}_t - c_t\|_2^2}{\max(\|c_t\|_2^2, \epsilon)}. \quad (3)$$

We average token losses within records and records across workers, with  $\epsilon$  the smallest positive normal FP32 value. Teacher-magnitude normalization prevents larger vectors from dominating. The normalized DFlash objective differs from raw linear least squares. The original fits use 4,096 math problem-and-solution conversations, capped at 192 tokens. Supervision is the source context vector. Only  $W$  is updated, with 52.43M/65.54M weights for 8B/14B targets. Appendix A gives dimensions, layers and fitting settings.

### 4.2 DRAFT WITH THE NEW TARGET’S FEATURES

The target caches prompt states, and the map supplies the frozen drafter’s context. The target verifies proposals in parallel, commits the matching prefix plus its next token, and crops the cache to the committed prefix. Mapped states feed the next cycle (Figure 6).

The standard greedy equivalence argument requires block verification to return the same token choices as single-token evaluation of each identical prefix (Leviathan et al., 2023). Finite-precision implementations can change those choices. We therefore measure task accuracy and token agreement separately from throughput. The map itself never supplies the scores used for target verification.

## 5 EXPERIMENTS

### 5.1 EXPERIMENTAL SETUP

**Models and runtime.** We retarget the released Qwen3-4B DFlash and DeepSpec EAGLE-3 drafters to Qwen3-8B and Qwen3-14B (Yang et al., 2025; DeepSeek-AI, 2026). DFlash uses block length 16. EAGLE-3 uses the supported chain procedure with seven draft positions. Native controls use released target-specific drafters within the same implementation and proposal setting. The main AR comparisons use four NVIDIA L40S GPUs, with one sequential request per worker. Historical block-size and isolated-memory studies use RTX 6000 Ada GPUs. We compare methods within runs on the same hardware. Serving systems such as vLLM and SGLang also optimize cache use and request execution (Kwon et al., 2023; Zheng et al., 2024). We measure one request per worker. Table 6 gives the full protocol.

**Workloads and data use.** We evaluate math, code and conversation workloads because speculative speed varies across tasks (Xia et al., 2024; Abramovich et al., 2026). MATH-500 supplies 500 questions (Lightman et al., 2024). The breadth suite contains 128 GSM8K questions, 164 HumanEval tasks, 378 MBPP tasks and 80 two-turn MT-Bench conversations (Hendrycks et al., 2021; Cobbe et al., 2021; Chen et al., 2021; Austin et al., 2021; Zheng et al., 2023). The 32-question development set informed interface and loss choices. Historical evaluations retain this exposure history. Fitting and evaluation problem texts have zero normalized exact matches. Appendix H quantifies related templates using token overlap. All completed workload cells in each reported suite are included.

**Controlled scaling protocol.** The original 4,096-record checkpoints remain fixed for the broad workload results. A fixed-1,024-update sweep varies distinct records from 64 to 4,096. A separate NuminaMath study uses nested 16–32,768-record sets, 1,024 feature-validation records and 8,192 batch-four updates per fit. Frozen feature caches are shared across matched linear and MLP fits. The longer-output capacity comparisons use 128 exposed MATH questions and a 2,048-token cap. The 14B capacity and adaptation screens use sixteen exposed questions and a 256-token cap. These studies have separate checkpoints and are not pooled with the broad evaluation.

**Quality and pairing.** We score the same math outputs used for the AR timing comparison with the pinned Qwen math scorer. Historical code comparisons use EvalPlus base and expanded tests (Liu et al., 2023). MT-Bench measures two-turn generation time, with each method conditioning its second turn on its own first answer. Its bootstrap unit is the conversation. Other confidence intervals resample matched requests and condition on the fitted checkpoint. Model loading and external task scoring sit outside the request timer.

### 5.2 ACCELERATION OVER PLAIN AUTOREGRESSIVE DECODING

Table 1 reports absolute end-to-end throughput and AR-relative speed. DFlash RelaySpec reaches 186.49 tokens/s at 8B and 116.88 at 14B, compared with AR at 37.37 and 22.88. EAGLE-3 RelaySpec reaches 84.92 and 59.18 tokens/s, compared with 36.15 and 22.88 for AR. The resulting throughput ratios are 4.99, 5.11, 2.35 and 2.59. Each paired interval is above one.

Table 1: MATH-500 end-to-end tokens/s, including prompt processing. Progress R is mean recorded relay progress per cycle. Ratios and intervals use the AR arm of that same run. Native-drafter comparisons are in Table 8.

| Drafter | Target | AR tok/s | Source tok/s | Relay tok/s | Relay / AR [95% CI] | Progress R |
|---------|--------|----------|--------------|-------------|---------------------|------------|
| DFlash  | 8B     | 37.37    | 121.81       | 186.49      | 4.99 [4.88, 5.10]   | 6.98       |
| DFlash  | 14B    | 22.88    | 92.15        | 116.88      | 5.11 [4.98, 5.23]   | 6.49       |
| EAGLE-3 | 8B     | 36.15    | 66.44        | 84.92       | 2.35 [2.32, 2.38]   | 4.33       |
| EAGLE-3 | 14B    | 22.88    | 53.96        | 59.18       | 2.59 [2.55, 2.63]   | 4.01       |

DFlash uses parallel proposals and the EAGLE-3 path drafts sequentially. These released-model comparisons also use different library versions, so cross-family speed does not isolate architecture.

### 5.3 NATIVE RETENTION AND SOURCE REMOVAL

RelaySpec retains 90.3% of native DFlash-8B throughput, reaching 186.49 versus 206.58 tokens/s. EAGLE-3 retains 90.2% at 8B and 89.1% at 14B (Table 8). These compare released drafters, not matched original training costs. Retained progress is lower than retained throughput because per-cycle work also differs: DFlash-8B advances 6.98 tokens per relay cycle versus 7.89 for native drafting.

Against source reuse, the relay’s throughput ratios are 1.531 and 1.268 for DFlash, and 1.278 and 1.097 for EAGLE-3, at 8B and 14B respectively. This comparison holds the inherited drafter fixed. Source computation is removed from each cycle, while retained target states supply the context at the scale tested in Section 5.7.

### 5.4 MATH ACCURACY AND OUTPUT AGREEMENT

Table 2 scores the exact outputs timed in Table 1. Observed AR-to-relay accuracy differences range from  $-0.2$  to  $+0.8$  percentage points, and each paired interval includes zero. These measured outcomes support the accuracy comparison without treating token equality as a substitute for task quality.

Table 2: MATH-500 accuracy in percent. Difference is RelaySpec minus AR in percentage points. Token matches count identical full output sequences.

| Drafter | Target | AR score | Relay score | Difference [95% CI] | Token matches |
|---------|--------|----------|-------------|---------------------|---------------|
| DFlash  | 8B     | 75.8     | 76.6        | +0.8 [-1.4, +3.0]   | 117/500       |
| DFlash  | 14B    | 79.2     | 79.2        | +0.0 [-2.2, +2.0]   | 113/500       |
| EAGLE-3 | 8B     | 75.8     | 75.6        | -0.2 [-2.4, +2.0]   | 135/500       |
| EAGLE-3 | 14B    | 79.2     | 79.4        | +0.2 [-1.8, +2.2]   | 106/500       |

BF16 AR and block verification can select different tokens on the same prefix. Eight selected DFlash traces show target-score changes across native drafting, source reuse and RelaySpec. A separate Qwen3-8B diagnostic yields 12/16 AR matches in BF16 and 16/16 in full FP32 (Appendix C). These diagnostics do not establish bitwise agreement generally.

### 5.5 SPEED ACROSS WORKLOADS

Figure 3 and Table 9 report the completed DFlash AR-paired breadth suite. RelaySpec is 1.85–3.69 times faster than AR across these eight cells. Code gains at 14B remain substantial relative to AR even where source reuse is faster than the relay. The useful comparison depends on the deployment choice: AR measures the acceleration obtained, while source reuse measures the benefit of removing its source.

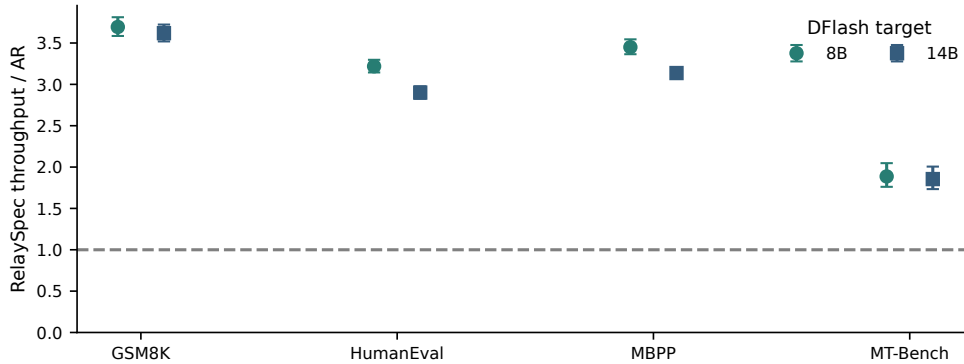


Figure 3: DFlash RelaySpec throughput relative to AR across workloads. Whiskers are paired 95% intervals. Chat intervals resample whole two-turn conversations. Full absolute values are in Table 9.

## 5.6 FITTING WORK AND DEPLOYMENT MEMORY

The selected maps have 52.43–65.54M weights. Fitting takes 85.6–118.5 seconds on four RTX 6000 Ada GPUs after model loading, including feature computation and optimization. Loading, export and deployment initialization are separate costs (Appendix I).

Separate EAGLE-3 processes measure memory after removing source layers. Peak allocated memory falls from 25.56 to 17.76 GiB at 8B and from 37.56 to 29.94 GiB at 14B. These are savings of 7.80 and 7.63 GiB. Source token embeddings and vocabulary projections required by a drafter remain part of deployment. The claim concerns the removed source transformer, not every tensor inherited from the source.

## 5.7 ABLATION STUDIES

### 5.7.1 HOW MUCH DATA AND MAPPER CAPACITY ARE NEEDED?

We separate distinct training examples from optimization work. With 8,192 batch-four updates held fixed, the NuminaMath sweep spans 16 to 32,768 distinct records. DFlash-8B throughput rises from 96.9 tokens/s at 16 examples to 192.8 at 512, then ranges from 196.9 to 198.4 at 2,048–32,768 (Figure 4). Thus 16 examples already give  $2.58\times$  AR throughput, and 512 retain 97.2% of the observed best. All eleven mappers score 105/128. These are single-seed development results, not a general minimum-data threshold. Appendix C retains the separate historical MATH-only study.

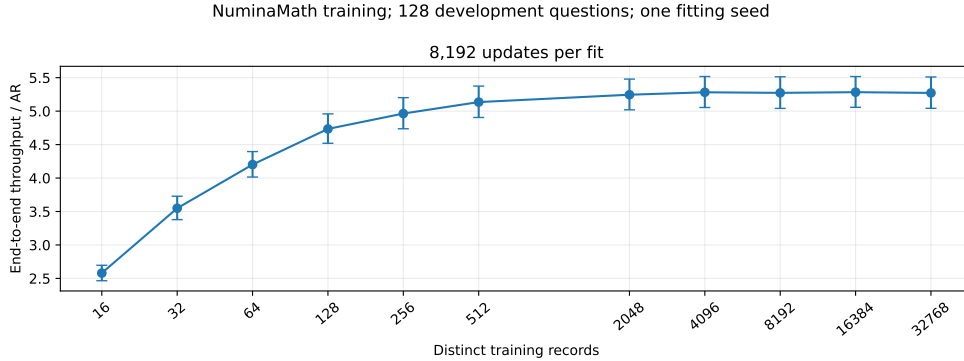


Figure 4: DFlash-8B data scaling on 128 development questions, with NuminaMath training and 8,192 updates per fit. All dataset sizes are measured with shared controls in one campaign. Paired 95% request intervals exclude fitting-seed variation and selection uncertainty.

The NuminaMath capacity study crosses 512/2,048 examples with dense, factored linear and GELU MLP maps over seven widths, using 8,192 updates per cell. This holds parameter count equal between each factored map and MLP. Validation error decreases with capacity and then flattens (Figure 5). On 128 questions at a 2,048-token cap, dense  $N = 512$  retains 97.78% [97.09, 98.47]% of  $N = 2048$  throughput. Linear-1024 at  $N = 512$  retains 95.00% [94.14, 95.85]% with 23.59M parameters instead of 52.43M. The latter interval crosses the 95% boundary. These are development comparisons with fixed fitting seeds.

Two-layer conditioning also transfers to native drafting. Frozen retargeted and native two-layer interfaces each cut projection parameters by 60%, retaining 97.63% [96.23, 99.03]% and 97.05% [95.93, 98.21]% throughput on disjoint 64-question GSM8K sets. Outputs match their respective full-interface references (Appendix C). Inherited native columns reduce calibration needs in matched development tests, but a fresh 16-record fit fails the fixed retention criterion (Tables 25 and 26).

EAGLE-3 repeats the dense data-count contrast on the same development questions, retaining 98.28% [97.59, 99.01]%. Its matched MLP remains slower than dense. Table 40 retains all eight maps.

**Confirmation with frozen checkpoints.** Before generating confirmation outputs, we fix the dense checkpoints and analysis criteria, then evaluate 256 GSM8K questions excluded from the recorded

development history (Table 3). Both families exceed the predeclared 95% speed-retention threshold. The two data sizes have identical per-question correctness within each family. Relative to AR, conservative accuracy-difference intervals are  $[-3.07, 3.07]$  points for DFlash and  $[-1.63, 3.14]$  for EAGLE-3. Neither establishes the one-point noninferiority margin.

Table 3: Frozen small-data confirmation on 256 GSM8K questions. Retention is a percentage relative to the 2,048-example mapper in the same family, with paired 95% intervals. Outputs are capped at 2,048 tokens.

| Family  | Examples | Tokens/s | Retention (95% CI) | Correct | At cap |
|---------|----------|----------|--------------------|---------|--------|
| DFlash  | 512      | 157.8    | 97.5 [96.9,98.1]   | 236/256 | 0      |
| DFlash  | 2,048    | 161.9    | 100 (reference)    | 236/256 | 0      |
| EAGLE-3 | 512      | 100.0    | 98.9 [98.1,99.7]   | 238/256 | 0      |
| EAGLE-3 | 2,048    | 101.1    | 100 (reference)    | 238/256 | 0      |

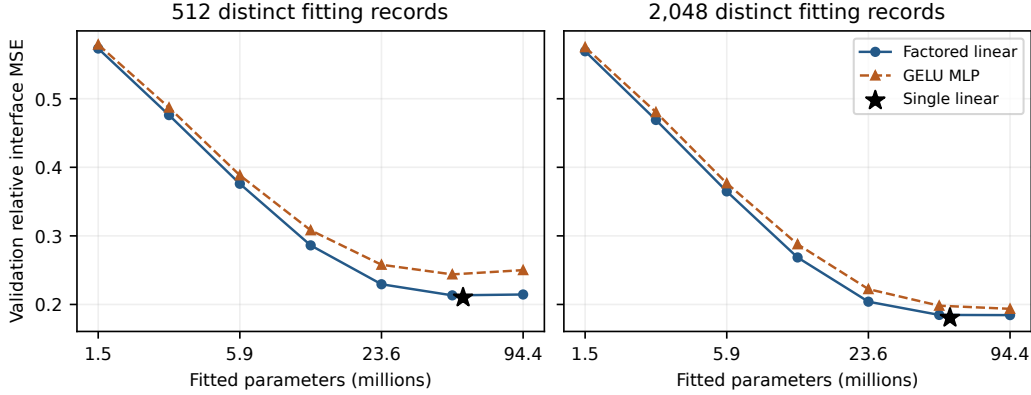


Figure 5: Complete primary capacity study at 8,192 updates per cell. Markers compare validation interface error, with one fitting seed. The curves do not establish decoding speed or answer quality.

**Better feature fitting need not mean faster decoding.** The 94.37M-parameter MLP-4096 at  $N = 2048$  reaches 190.15 tokens/s, below dense. Extending the width-512 MLP from 8,192 to 32,768 updates on the same 2,048 records raises throughput only from 154.39 to 157.68. All tested mappers score 105/128 at this cap, versus AR’s 104/128, which does not establish a one-percentage-point noninferiority margin. The regularization study finds no matched endpoint benefit from its L2 settings and only small weight-decay gains in feature validation. The dense linear map has the highest mapper throughput in both completed 8B long-output capacity comparisons. Thus its simplicity is supported by measured alternatives, including equal-capacity nonlinear maps, rather than assumed from parameter count. All outcomes remain in the appendix.

**Replication and workload dependence.** The complete 14B grids repeat the small-data/capacity comparison in both drafter families. DFlash’s wide MLP fits its training features better than dense but has worse validation error and lower capped decoding throughput. EAGLE-3’s late degradation and wide-linear loss spike are reduced by lower-rate follow-ups, which improve both dense and MLP decoding in a small paired screen. These results delimit the optimization recipe rather than prove a function-class disadvantage (Appendix C.6). On the 8B capped long-output development set, dense  $N = 512$  retains 97.4–97.9% of dense  $N = 2048$  throughput across MATH difficulty groups. Very long inputs have too few examples for a reliable generalization claim.

**Training composition changes the speed tradeoff.** Replacing half of 2,048 math records with general instructions reduces dense general-domain feature error by 32.9% but raises math error by 10.2%. Code and chat throughput improve by 2.9% and 4.9%, while MATH throughput falls 1.2%. A second seed repeats this pattern. Matched arms generate identical tokens on all 160 development



requests. For two-layer maps, a separate 16-conversation extension gives a 7.5% dialogue gain at both fitting seeds, compared with 0.3–1.1% for native interface students (Appendix F).

### 5.7.2 FEATURE FITTING WINS THE MATCHED ADAPTATION COMPARISONS

Feature regression achieves the highest throughput at every tested warm budget (Table 4). The comparison fixes 512 training records, charges CE and LoRA for their initial mapper, and gives each an equal three-rate development screen. Across all three budgets, CE retains 73.2–76.6% and the two LoRA fits retain 79.4–82.7% of feature throughput. Every paired 95% interval lies below parity. This supports feature regression as the best tested adaptation under these budgets. The sixteen-question, 256-token screen does not establish full-answer quality or the ranking of other objectives. Cache preparation and export remain separate costs (Appendix C.7).

Table 4: Decoding tokens/s after matched warm-training budgets, using one common DFlash-8B runtime and the same 512 records. Columns are nominal training seconds. Bold marks the best tested adaptation in each column. All twelve fits and paired intervals appear in Table 45.

| Adaptation    | 22.9 s        | 91.4 s        | 365.6 s       |
|---------------|---------------|---------------|---------------|
| Feature MSE   | <b>145.05</b> | <b>144.80</b> | <b>144.55</b> |
| Connector CE  | 111.04        | 109.60        | 105.79        |
| LoRA32 (1729) | 119.30        | 115.02        | 118.49        |
| LoRA32 (1730) | 119.52        | 115.49        | 119.55        |

**Public-runtime baselines.** Table 5 places the completed public baselines beside RelaySpec. The relay has the highest measured throughput, but the separate runtime controls prevent an algorithm-only ranking. Table 47 distinguishes their adaptation mechanisms.

Table 5: Public-runtime comparison on 128 MATH development questions, 2,048-token cap. Correct counts are out of 128, with each system’s AR in parentheses. Precision, attention and libraries differ, so speedups use own-runtime AR. Settings and paired quality uncertainty appear in Appendix D.

| System                           | Tok/s  | Own AR tok/s | Speedup [95% CI]  | Correct (AR) |
|----------------------------------|--------|--------------|-------------------|--------------|
| RelaySpec ( $N = 512$ )          | 193.68 | 37.72        | 5.14 [4.90, 5.37] | 105 (104)    |
| PARD                             | 105.63 | 31.67        | 3.34 [3.22, 3.45] | 105 (103)    |
| SD <sup>2</sup> (frozen drafter) | 19.92  | 12.48        | 1.60 [1.56, 1.63] | 103 (103)    |

## 6 CONCLUSION

RelaySpec makes an existing feature-conditioned drafter useful with a larger verifier by learning one replacement input map. Its value is the combination of retained drafting behavior, removed source-transformer cost and little additional fitting data. Dense linear mapping is a competitive choice after testing nonlinear capacity, longer optimization, regularization and token-prediction adaptation. The interface also exposes which target features sustain drafting: late-layer selection cuts mapper parameters by 60% while retaining 97.6% throughput in a fixed comparison. A native two-layer student retains 97.1% throughput with 60% fewer projection parameters in a separate fixed comparison. These findings motivate adapting and inspecting the inherited interface before investing in a new drafter.

These conclusions concern shared-vocabulary Qwen3 models, greedy decoding and one request per worker. Code workloads can favor source reuse, and EAGLE-3 optimization depends on the recipe. Frozen GSM8K confirmation supports speed retention but not tight accuracy noninferiority. Development exposure, possible shared math templates and limited fitting seeds restrict generalization. The 14B capacity and adaptation screens remain capped comparisons, and public-runtime baselines do not isolate an algorithm-only ranking.

## AI USE STATEMENT

Generative AI assisted with implementation, code review, literature search, analysis and manuscript editing. Numerical tables and plots are generated from recorded artifacts with checked method, request and scorer identity. Scientific interpretation and submission decisions remain the authors' responsibility.

## REPRODUCIBILITY STATEMENT

The appendix specifies model identifiers, feature layers, fitting settings, measurement boundaries, complete workload results and data checks. The accompanying implementation includes configurations, request records, scorer provenance and scripts that regenerate the reported assets.

## REFERENCES

- Talor Abramovich, Maor Ashkenazi, Izzy Putterman, Benjamin Chislett, Tiyasa Mitra, Bitu Darvish Rouhani, Ran Zilberstein, and Yonatan Geifman. SPEED-Bench: A unified and diverse benchmark for speculative decoding. In *43rd International Conference on Machine Learning*, 2026. URL <https://arxiv.org/abs/2604.09557>. Accepted paper.
- Zihao An, Huajun Bai, Ziqiong Liu, Dong Li, and Emad Barsoum. PARD: Accelerating LLM inference with low-cost PARallel draft model adaptation. In *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=XbOyv7iVGL>.
- Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, and Charles Sutton. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*, 2021. URL <https://arxiv.org/abs/2108.07732>.
- Yamini Bansal, Preetum Nakkiran, and Boaz Barak. Revisiting model stitching to compare neural representations. In *Advances in Neural Information Processing Systems 34*, 2021. URL <https://papers.nips.cc/paper/2021/hash/01ded4259d101feb739b06c399e9cd9c-Abstract.html>.
- Frédéric Berdoz, Peer Rheinboldt, and Roger Wattenhofer. Steering pretrained drafters during speculative decoding. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pp. 30067–30075, 2026. doi: 10.1609/aaai.v40i36.40255. URL <https://ojs.aaai.org/index.php/AAAI/article/view/40255>.
- Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D. Lee, Deming Chen, and Tri Dao. Medusa: Simple LLM inference acceleration framework with multiple decoding heads. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. URL <https://proceedings.mlr.press/v235/cai24b.html>.
- Jian Chen, Yesheng Liang, and Zhijian Liu. DFlash: Block diffusion for flash speculative decoding. In *43rd International Conference on Machine Learning*, 2026. URL <https://arxiv.org/abs/2602.06036>. Accepted paper.
- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021. URL <https://arxiv.org/abs/2107.03374>.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. URL <https://arxiv.org/abs/2110.14168>.
- DeepSeek-AI. DeepSpec: A full-stack codebase for training and evaluating speculative decoding draft models. Official open-source software, 2026. URL <https://github.com/deepseek-ai/DeepSpec>.

- Mostafa Elhoushi, Akshat Shrivastava, Diana Liskovich, Basil Hosmer, Bram Wasti, Liangzhen Lai, Anas Mahmoud, Bilge Acun, Saurabh Agarwal, Ahmed Roman, Ahmed Aly, Beidi Chen, and Carole-Jean Wu. LayerSkip: Enabling Early Exit Inference and Self-Speculative Decoding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024. URL <https://aclanthology.org/2024.acl-long.681/>.
- Yichao Fu, Peter Bailis, Ion Stoica, and Hao Zhang. Break the Sequential Dependency of LLM Inference Using Lookahead Decoding. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. URL <https://proceedings.mlr.press/v235/fu24a.html>.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *Advances in Neural Information Processing Systems 34, Datasets and Benchmarks Track*, 2021. URL <https://openreview.net/forum?id=7Bywt2mQsCe>.
- Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-Efficient Transfer Learning for NLP. In *Proceedings of the 36th International Conference on Machine Learning*, 2019. URL <https://proceedings.mlr.press/v97/houlsby19a.html>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations*, 2022. URL <https://www.microsoft.com/en-us/research/publication/lora-low-rank-adaptation-of-large-language-models/>.
- Shijing Hu, Jingyang Li, Xingyu Xie, Zhihui Lu, Kim-Chuan Toh, and Pan Zhou. GRIFFIN: Effective token alignment for faster speculative decoding. In *Advances in Neural Information Processing Systems*, 2025. URL [https://proceedings.neurips.cc/paper\\_files/paper/2025/hash/b6e67ae290635d0874c4cb43ba2a2cfb-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2025/hash/b6e67ae290635d0874c4cb43ba2a2cfb-Abstract-Conference.html).
- Feiye Huo, Jianchao Tan, Jiahao Liu, Zixu Jiang, Jiacheng Li, Jingang Wang, Xunliang Cai, and Shengli Sun. RepSpec: Structural re-parameterized draft model training for speculative decoding. In *International Conference on Learning Representations*, 2026. URL [https://proceedings.iclr.cc/paper\\_files/paper/2026/hash/d70ea003729b440b89a2f958a5554c1f-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2026/hash/d70ea003729b440b89a2f958a5554c1f-Abstract-Conference.html).
- Haoyun Jiang, Junqi He, Feng Hong, Xinlong Yang, Jianwei Zhang, Zheng Li, Zhengyang Zhuge, Zhiyong Chen, Bo Han, Junyang Lin, and Jiangchao Yao. TriSpec: Ternary speculative decoding via lightweight proxy verification. *arXiv preprint arXiv:2601.23180*, 2026. doi: 10.48550/arXiv.2601.23180. URL <https://arxiv.org/abs/2601.23180>.
- Tanishq Kumar, Tri Dao, and Avner May. Speculative speculative decoding. In *International Conference on Learning Representations*, 2026. URL [https://proceedings.iclr.cc/paper\\_files/paper/2026/hash/1b96f01343ff10150e6719eb163e1536-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2026/hash/1b96f01343ff10150e6719eb163e1536-Abstract-Conference.html).
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient Memory Management for Large Language Model Serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, 2023. URL <https://doi.org/10.1145/3600006.3613165>.
- Haodi Lei, Yafu Li, Haoran Zhang, Shunkai Zhang, Qianjia Cheng, Xiaoye Qu, Ganqu Cui, Bowen Zhou, Ning Ding, Yun Luo, and Yu Cheng. Draft-OPD: On-policy distillation for speculative draft models. *arXiv preprint arXiv:2605.29343*, 2026. URL <https://arxiv.org/abs/2605.29343>.
- Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 19274–19286, 2023. URL <https://proceedings.mlr.press/v202/leviathan23a.html>.

- Yuhui Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. EAGLE: Speculative Sampling Requires Rethinking Feature Uncertainty. In *Proceedings of the 41st International Conference on Machine Learning*, 2024a. URL <https://proceedings.mlr.press/v235/li24bt.html>.
- Yuhui Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. EAGLE-2: Faster Inference of Language Models with Dynamic Draft Trees. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024b. URL <https://aclanthology.org/2024.emnlp-main.422/>.
- Yuhui Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. EAGLE-3: Scaling up inference acceleration of large language models via training-time test. In *Advances in Neural Information Processing Systems 38*, 2025. URL [https://proceedings.neurips.cc/paper\\_files/paper/2025/hash/c7b5a35ea98b62512a869c19ea7b03cb-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2025/hash/c7b5a35ea98b62512a869c19ea7b03cb-Abstract-Conference.html).
- Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International Conference on Learning Representations*, 2024. URL [https://proceedings.iclr.cc/paper\\_files/paper/2024/hash/aca97732e30bcf1303bc22ac3924fd16-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2024/hash/aca97732e30bcf1303bc22ac3924fd16-Abstract-Conference.html).
- Jiawei Liu, Chunqiu Steven Xia, Yuyao Wang, and Lingming Zhang. Is your code generated by ChatGPT really correct? rigorous evaluation of large language models for code generation. In *Advances in Neural Information Processing Systems 36*, 2023. URL [https://proceedings.neurips.cc/paper\\_files/paper/2023/hash/43e9d647ccd3e4b7b5baab53f0368686-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2023/hash/43e9d647ccd3e4b7b5baab53f0368686-Abstract-Conference.html).
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL <https://arxiv.org/abs/1711.05101>.
- Xupeng Miao, Gabriele Oliaro, Zhihao Zhang, Xinhao Cheng, Zeyu Wang, Zhengxin Zhang, Rae Ying Yee Wong, Alan Zhu, Lijie Yang, Xiaoxiang Shi, Chunan Shi, Zhuoming Chen, Daiyaan Arfeen, Reyna Abhyankar, and Zhihao Jia. SpecInfer: Accelerating Large Language Model Serving with Tree-based Speculative Inference and Verification. In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 3*, 2024. URL <https://arxiv.org/abs/2305.09781>.
- Ramchalam Kinattinkara Ramakrishnan, Zhaocong Yuan, Shaojie Zhuo, Chen Feng, Yicheng Lin, Chenzheng Su, and Xiaopeng Zhang. OmniDraft: A cross-vocabulary, online adaptive drafter for on-device speculative decoding. In *Advances in Neural Information Processing Systems*, 2025. URL [https://proceedings.neurips.cc/paper\\_files/paper/2025/hash/3c2fe1417eed1c6ff9acf169617981ea-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2025/hash/3c2fe1417eed1c6ff9acf169617981ea-Abstract-Conference.html).
- Damian Smith, Harvey Mannerling, and Antonia Marcu. Functional alignment can mislead: Examining model stitching. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 55972–55998, 2025. URL <https://proceedings.mlr.press/v267/smith25a.html>.
- Mitchell Stern, Noam Shazeer, and Jakob Uszkoreit. Blockwise Parallel Decoding for Deep Autoregressive Models. In *Advances in Neural Information Processing Systems 31*, 2018. URL <https://proceedings.neurips.cc/paper/2018/hash/c4127b9194fe8562c64dc0f5bf2c93bc-Abstract.html>.
- Nadav Timor, Jonathan Mamou, Daniel Korat, Moshe Berchansky, Gaurav Jain, Oren Pereg, Moshe Wasserblat, and David Harel. Accelerating LLM inference with lossless speculative decoding algorithms for heterogeneous vocabularies. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025a. URL <https://proceedings.mlr.press/v267/timor25a.html>.
- Nadav Timor, Jonathan Mamou, Daniel Korat, Moshe Berchansky, Oren Pereg, Moshe Wasserblat, Tomer Galanti, Michal Gordon-Kiwkowitz, and David Harel. Distributed speculative inference: Speculation parallelism for provably faster lossless language model inference. In *International Conference on Learning Representations*,

- 2025b. URL [https://proceedings.iclr.cc/paper\\_files/paper/2025/hash/b36554b97da741b1c48c9de05c73993e-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2025/hash/b36554b97da741b1c48c9de05c73993e-Abstract-Conference.html).
- Jikai Wang, Yi Su, Juntao Li, Qingrong Xia, Zi Ye, Xinyu Duan, Zhefeng Wang, and Min Zhang. OPT-Tree: Speculative Decoding with Adaptive Draft Tree Structure. *Transactions of the Association for Computational Linguistics*, 2025a. URL <https://aclanthology.org/2025.tacl-1.8/>.
- Xin Wang, Yu Zheng, Zhongwei Wan, and Mi Zhang. SVD-LLM: Truncation-aware singular value decomposition for large language model compression. In *International Conference on Learning Representations*, 2025b. URL <https://arxiv.org/abs/2403.07378>.
- Heming Xia, Tao Ge, Peiyi Wang, Si-Qing Chen, Furu Wei, and Zhifang Sui. Speculative Decoding: Exploiting Speculative Execution for Accelerating Seq2seq Generation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. URL <https://aclanthology.org/2023.findings-emnlp.257/>.
- Heming Xia, Zhe Yang, Qingxiu Dong, Peiyi Wang, Yongqi Li, Tao Ge, Tianyu Liu, Wenjie Li, and Zhifang Sui. Unlocking Efficiency in Large Language Model Inference: A Comprehensive Survey of Speculative Decoding. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024. URL <https://aclanthology.org/2024.findings-acl.456/>.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. URL <https://arxiv.org/abs/2505.09388>.
- Jiayi Yuan, Hao Li, Xinheng Ding, Wenya Xie, Yu-Jhe Li, Wentian Zhao, Kun Wan, Jing Shi, Xia Hu, and Zirui Liu. Understanding and mitigating numerical sources of nondeterminism in LLM inference. *arXiv preprint arXiv:2506.09501*, 2025. URL <https://arxiv.org/abs/2506.09501>.
- Zhihang Yuan, Yuzhang Shang, Yue Song, Dawei Yang, Qiang Wu, Yan Yan, and Guangyu Sun. ASVD: Activation-aware singular value decomposition for compressing large language models. *arXiv preprint arXiv:2312.05821*, 2023. URL <https://arxiv.org/abs/2312.05821>.
- Biao Zhang and Rico Sennrich. Root Mean Square Layer Normalization. In *Advances in Neural Information Processing Systems 32*, 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/1e8a19426224ca89e83cef47f1e7f53b-Abstract.html>.
- Jun Zhang, Jue Wang, Huan Li, Lidan Shou, Ke Chen, Gang Chen, and Sharad Mehrotra. Draft & Verify: Lossless Large Language Model Acceleration via Self-Speculative Decoding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024. URL <https://aclanthology.org/2024.acl-long.607/>.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems 36, Datasets and Benchmarks Track*, 2023. URL [https://proceedings.neurips.cc/paper\\_files/paper/2023/hash/91f18a1287b398d378ef22505bf41832-Abstract-Datasets\\_and\\_Benchmarks.html](https://proceedings.neurips.cc/paper_files/paper/2023/hash/91f18a1287b398d378ef22505bf41832-Abstract-Datasets_and_Benchmarks.html).
- Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Chuyue Sun, Jeff Huang, Cody Hao Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph E. Gonzalez, Clark Barrett, and Ying Sheng. SGLang: Efficient Execution of Structured Language Model Programs. In *Advances in Neural Information Processing Systems 37*, 2024. URL [https://proceedings.neurips.cc/paper\\_files/paper/2024/hash/724be4472168f31balc9ac630f15dec8-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2024/hash/724be4472168f31balc9ac630f15dec8-Abstract-Conference.html).
- Xiandong Zou, Jianshu Li, Jing Huang, and Pan Zhou. Variational speculative decoding: Rethinking draft training from token likelihood to sequence acceptance. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. URL [https://ink.library.smu.edu.sg/sis\\_research/11190/](https://ink.library.smu.edu.sg/sis_research/11190/).

## APPENDIX

The appendix gives implementation details (Appendix A), complete workload results (Appendix B), fitting studies (Appendix C), acceptance measurements (Appendix E), data checks (Appendix H) and fitting-time and memory measurements (Appendix I).

### A IMPLEMENTATION AND MEASUREMENT DETAILS

Table 6: Evaluation protocol. Main fitting choices remain fixed across workloads. Full identifiers and runtime versions are in Appendix A.

| Component        | Setting                                                                                            |
|------------------|----------------------------------------------------------------------------------------------------|
| Primary baseline | Plain greedy AR with the same target, prompt and runtime                                           |
| Other controls   | Native target-specific drafter and optimized source reuse                                          |
| Generation       | BF16, SDPA attention, thinking disabled, maximum 2,048 new tokens                                  |
| Relay fitting    | 4,096 records, 192-token cap, 1,024 AdamW updates (Loshchilov & Hutter, 2019), four records/update |
| Optimizer        | Learning rate $6 \times 10^{-4}$ , zero weight decay, gradient clipping 1.0                        |
| Timing           | Synchronized request timer, warmup excluded, rotated method order                                  |
| Uncertainty      | 10,000 paired resamples, whole conversations for two-turn chat                                     |

**Released models.** The source is Qwen/Qwen3-4B. Targets are Qwen/Qwen3-8B and Qwen/Qwen3-14B. The source DFlash is z-lab/Qwen3-4B-DFlash-b16, and its native 8B control is z-lab/Qwen3-8B-DFlash-b16. EAGLE-3 uses DeepSpec’s eagle3\_qwen3\_4b\_ttt7, eagle3\_qwen3\_8b\_ttt7 and eagle3\_qwen3\_14b\_ttt7 releases. All model and upstream revisions are fixed in the accompanying configurations.

**Features and dimensions.** Layer indices count transformer blocks from zero. Source and Qwen3-8B states come from blocks [1, 9, 17, 25, 33]. Qwen3-14B states come from [1, 10, 19, 28, 37]. Joining five 4,096-wide states gives a 20,480-wide 8B input. Joining five 5,120-wide states gives a 25,600-wide 14B input. Both drafter interfaces have width 2,560. The matrix dimensions are therefore  $2,560 \times 20,480$  and  $2,560 \times 25,600$ . The DFlash input normalization has no trainable gain or bias. The released output normalization remains frozen.

**Training.** Four workers average record losses at each update. Seed 1729 fixes the selected fit. AdamW uses learning rate 0.0006, zero weight decay and gradient clipping at 1.0. Computation uses BF16, with the feature loss computed in FP32. Each of 1,024 updates processes four records, each capped at 192 rendered tokens. The same map remains fixed across tasks.

**Runtime and timing.** The main AR runs use Python 3.12.13 and PyTorch 2.9.1 with CUDA 12.8. DFlash uses Transformers 5.3.0. EAGLE-3 uses the pinned 5.10.2 overlay. Each worker owns a full target replica on one L40S GPU. Four workers process independent requests rather than splitting one model over GPUs. One warmup per method precedes measured requests. Method order rotates across requests. The synchronized timer includes prompt processing, drafting, verification, feature updates and runtime bookkeeping. The historical block and memory experiments use RTX 6000 Ada GPUs.

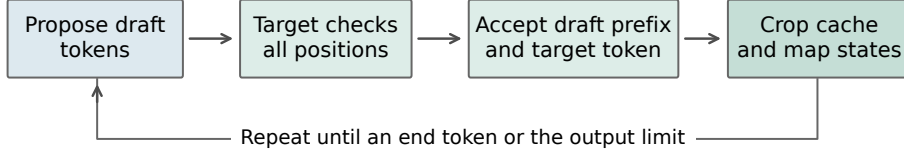


Figure 6: One generation cycle. RelaySpec changes the drafter’s context. The target retains control over committed tokens and the shared decoder retains responsibility for positions, stopping and cache updates.

**Stopping and conversation history.** Generation stops at the first end token or the output cap. Capped requests remain in both timing and quality aggregates. Each MT-Bench method uses its own first-turn output as second-turn context. Two turns form one resampling unit. The breadth run has 750 initial records and 830 timed requests after expanding 80 conversations to two turns.

**Aggregation.** Each method’s throughput is the sum of its output-token counts divided by the sum of its request times. Each of 10,000 resamples recomputes this ratio from matched requests, using bootstrap seed 1729. Confidence bounds are the central 95% of resampled estimates. The request-time ratio uses summed times alone. Table 7 reports both measurements and output lengths.

Table 7: Additional MATH-500 measurements. A is AR, S is source reuse and R is RelaySpec. Time ratio divides summed AR time by summed relay time.

| Drafter | Target | Mean tokens AR / R | Time ratio AR / R | Throughput R / S | Progress S / R |
|---------|--------|--------------------|-------------------|------------------|----------------|
| DFlash  | 8B     | 783.22 / 778.81    | 5.018             | 1.531            | 7.66 / 6.98    |
| DFlash  | 14B    | 777.66 / 772.33    | 5.143             | 1.268            | 7.44 / 6.49    |
| EAGLE-3 | 8B     | 783.22 / 767.79    | 2.397             | 1.278            | 5.15 / 4.33    |
| EAGLE-3 | 14B    | 777.66 / 766.20    | 2.625             | 1.097            | 5.07 / 4.01    |

## B COMPLETE WORKLOAD RESULTS

Table 8: Native throughput retained and mean recorded progress per cycle. N and R denote native drafting and RelaySpec. Progress includes the implementation’s target-supplied token.

| Drafter | Target | Native tok/s | Relay tok/s | Retained [95% CI]  | Progress N | Progress R |
|---------|--------|--------------|-------------|--------------------|------------|------------|
| DFlash  | 8B     | 206.58       | 186.49      | 90.3% [89.8, 90.8] | 7.89       | 6.98       |
| EAGLE-3 | 8B     | 94.12        | 84.92       | 90.2% [89.8, 90.7] | 5.40       | 4.33       |
| EAGLE-3 | 14B    | 66.39        | 59.18       | 89.1% [88.6, 89.7] | 5.28       | 4.01       |

Table 9: Complete DFlash breadth comparisons with paired AR. Ratios use end-to-end throughput. Every planned task and target cell is included.

| Target | Task      | Requests | AR tok/s | Relay tok/s | Relay / AR [95% CI] | Relay / source |
|--------|-----------|----------|----------|-------------|---------------------|----------------|
| 8B     | GSM8K     | 128      | 37.60    | 138.89      | 3.69 [3.59, 3.81]   | 1.456          |
| 8B     | HumanEval | 164      | 37.55    | 120.87      | 3.22 [3.14, 3.30]   | 1.244          |
| 8B     | MBPP      | 378      | 37.49    | 129.35      | 3.45 [3.36, 3.54]   | 1.320          |
| 8B     | MT-Bench  | 160      | 37.47    | 70.69       | 1.89 [1.76, 2.05]   | 1.387          |
| 14B    | GSM8K     | 128      | 26.80    | 96.98       | 3.62 [3.52, 3.72]   | 1.104          |
| 14B    | HumanEval | 164      | 26.73    | 77.54       | 2.90 [2.83, 2.98]   | 0.890          |
| 14B    | MBPP      | 378      | 26.72    | 83.82       | 3.14 [3.07, 3.21]   | 0.956          |
| 14B    | MT-Bench  | 160      | 26.55    | 49.26       | 1.85 [1.73, 2.01]   | 1.086          |

The earlier fixed-map source comparisons in Table 10 provide all four workloads for both drafter families on RTX 6000 Ada. They are a separate paired experiment with their own denominator. The EAGLE-3 14B code cells show the reduced margin when lower accepted progress outweighs source-removal savings. This operating range complements the AR acceleration measured in the main comparison.

Table 10: Historical paired source-reuse breadth measurements on RTX 6000 Ada. Ratios compare relay with source throughput within these runs. They are not normalized by an AR arm from another experiment.

| Drafter | Target | Task      | Requests | Source tok/s | Relay tok/s | Ratio [95% interval]    |
|---------|--------|-----------|----------|--------------|-------------|-------------------------|
| DFlash  | 8B     | GSM8K     | 128      | 116.87       | 166.39      | 1.4237 [1.4040, 1.4445] |
| DFlash  | 8B     | HumanEval | 164      | 118.84       | 145.00      | 1.2201 [1.2017, 1.2383] |
| DFlash  | 8B     | MBPP      | 378      | 117.77       | 151.55      | 1.2868 [1.2732, 1.3001] |
| DFlash  | 8B     | MT-Bench  | 160      | 60.85        | 81.62       | 1.3412 [1.3183, 1.3639] |
| DFlash  | 14B    | GSM8K     | 128      | 88.04        | 97.07       | 1.1025 [1.0836, 1.1214] |
| DFlash  | 14B    | HumanEval | 164      | 87.34        | 77.61       | 0.8886 [0.8741, 0.9038] |
| DFlash  | 14B    | MBPP      | 378      | 88.02        | 83.98       | 0.9541 [0.9416, 0.9663] |
| DFlash  | 14B    | MT-Bench  | 160      | 45.32        | 49.30       | 1.0877 [1.0653, 1.1083] |
| EAGLE-3 | 8B     | GSM8K     | 128      | 85.99        | 108.45      | 1.2613 [1.2437, 1.2789] |
| EAGLE-3 | 8B     | HumanEval | 164      | 71.75        | 81.10       | 1.1304 [1.1172, 1.1434] |
| EAGLE-3 | 8B     | MBPP      | 378      | 69.85        | 82.10       | 1.1753 [1.1656, 1.1850] |
| EAGLE-3 | 8B     | MT-Bench  | 160      | 42.27        | 49.45       | 1.1698 [1.1501, 1.1906] |
| EAGLE-3 | 14B    | GSM8K     | 128      | 67.81        | 73.41       | 1.0826 [1.0675, 1.0979] |
| EAGLE-3 | 14B    | HumanEval | 164      | 55.38        | 50.58       | 0.9132 [0.9011, 0.9255] |
| EAGLE-3 | 14B    | MBPP      | 378      | 54.55        | 50.49       | 0.9256 [0.9151, 0.9361] |
| EAGLE-3 | 14B    | MT-Bench  | 160      | 33.83        | 34.57       | 1.0221 [0.9999, 1.0449] |

Table 11: Task scores for the historical fixed-map breadth runs. Source reuse and RelaySpec share each reported score. Code columns are single-program pass percentages under base and expanded EvalPlus tests.

| Drafter | Target | GSM8K | HumanEval | HumanEval+ | MBPP  | MBPP+ |
|---------|--------|-------|-----------|------------|-------|-------|
| DFlash  | 8B     | 93.75 | 85.98     | 80.49      | 84.13 | 73.02 |
| DFlash  | 14B    | 94.53 | 89.02     | 83.54      | 88.36 | 75.13 |
| EAGLE-3 | 8B     | 92.97 | 87.20     | 82.32      | 83.07 | 71.96 |
| EAGLE-3 | 14B    | 94.53 | 90.24     | 85.98      | 88.89 | 74.87 |

Across these historical math/code comparisons, all 4,680 paired task outcomes agree between source reuse and RelaySpec. This count is four model/drafter pairs times  $500 + 128 + 164 + 378 = 1,170$  problems. Expanded code tests evaluate the same programs and do not add independent examples. For the current AR-paired DFlash GSM8K runs, the scorer gives 120/128 correct for all arms at 8B and 121/128 at 14B. Code scores above refer specifically to the historical outputs, preserving scorer/run identity.

**AR-paired EAGLE-3 code quality.** We additionally score the saved programs from the EAGLE-3 AR-paired breadth runs with official EvalPlus base and expanded tests. Table 12 compares one program per task against AR from the same run. At 8B, RelaySpec passes four fewer MBPP tasks under both suites. At 14B, it passes two more. HumanEval expanded-test totals match AR at both target sizes, although individual task outcomes differ. RelaySpec and source reuse have identical per-task outcomes in all eight cells. The paired intervals cover zero. This does not establish equality or noninferiority. These retrospective comparisons use one fitted map per setting and do not include fitting-seed uncertainty. Conversation quality remains unmeasured.



Table 12: EAGLE-3 AR-paired code quality. Counts are passed tasks. The intervals resample paired tasks. Expanded tests reuse the same programs and do not add independent samples.

| Target/task   | Suite | AR      | Relay   | $\Delta$ (pp) | Paired 95% interval |
|---------------|-------|---------|---------|---------------|---------------------|
| 8B HumanEval  | base  | 143/164 | 141/164 | -1.22         | $[-3.66, +1.22]$    |
| 8B HumanEval  | plus  | 133/164 | 133/164 | +0.00         | $[-3.05, +3.05]$    |
| 8B MBPP       | base  | 318/378 | 314/378 | -1.06         | $[-2.91, +0.79]$    |
| 8B MBPP       | plus  | 276/378 | 272/378 | -1.06         | $[-2.65, +0.53]$    |
| 14B HumanEval | base  | 143/164 | 143/164 | +0.00         | $[-2.44, +2.44]$    |
| 14B HumanEval | plus  | 134/164 | 134/164 | +0.00         | $[-2.44, +2.44]$    |
| 14B MBPP      | base  | 338/378 | 340/378 | +0.53         | $[-1.32, +2.38]$    |
| 14B MBPP      | plus  | 283/378 | 285/378 | +0.53         | $[-1.06, +2.12]$    |

## C FITTING AND INTERFACE STUDIES

Table 13: DFlash-8B block-size ablation on 64 questions, a 512-token cap and RTX 6000 Ada hardware. AR is measured in each run.

| Block | AR tok/s | Native draft tok/s | Relay tok/s | Relay / AR | Progress / cycle |
|-------|----------|--------------------|-------------|------------|------------------|
| 8     | 44.68    | 175.12             | 164.90      | 3.691      | 4.98             |
| 16    | 44.67    | 229.87             | 210.15      | 4.704      | 6.48             |
| 32    | 44.76    | 207.70             | 131.36      | 2.935      | 4.14             |

**Distinct records at fixed fitting work.** A separate controlled DFlash-8B study holds fitting at 1,024 updates and four records per update while varying distinct records from 64 to 4,096. Smaller sets repeat in the same ordered schedule. Table 14 reports end-to-end throughput on the same 128 development-exposed MATH questions. Of the measured settings, 512 is the smallest within 5% of the best observed mapper throughput. The 256-record setting retains 91.4% of that observed best. This is a descriptive, single-seed boundary, not a minimum across settings or a confirmed optimum. The 512–4,096 settings form an unresolved plateau. Selecting the reference from the same evaluations does not provide independent confirmation. Every relay scores 105/128, versus AR’s 104/128.

Table 14: Distinct-data study at 4,096 presentations for every row. Native retention uses each run’s paired native-drafter control. Best retention uses the observed best relay across these settings. All results condition on one fitting seed.

| Records | Tokens/s | $\times$ AR | Native retained | Best retained | Correct |
|---------|----------|-------------|-----------------|---------------|---------|
| 64      | 134.39   | 3.55        | 65.7%           | 72.8%         | 105/128 |
| 128     | 154.56   | 4.10        | 75.5%           | 83.7%         | 105/128 |
| 256     | 168.76   | 4.51        | 82.6%           | 91.4%         | 105/128 |
| 512     | 181.91   | 4.84        | 88.9%           | 98.5%         | 105/128 |
| 1,024   | 184.55   | 4.92        | 90.3%           | 99.9%         | 105/128 |
| 2,048   | 183.81   | 4.88        | 89.8%           | 99.5%         | 105/128 |
| 4,096   | 184.64   | 4.90        | 90.2%           | 100.0%        | 105/128 |

**Continuous optimization.** We also save checkpoints from one uninterrupted 4,096-record fit at 128, 256, 512, 1,024 and 2,048 updates. They have seen respectively 512, 1,024, 2,048, 4,096 and 4,096 distinct records. The final point therefore measures a second pass, not 8,192 distinct examples. End-to-end AR-relative throughput increases from 3.65 to 5.02 across this trajectory. All five checkpoints score 105/128. Unlike the earlier separately fitted budget configurations, these checkpoints share one optimization trajectory. Figure 7 separates this experiment from the fixed-work data study.

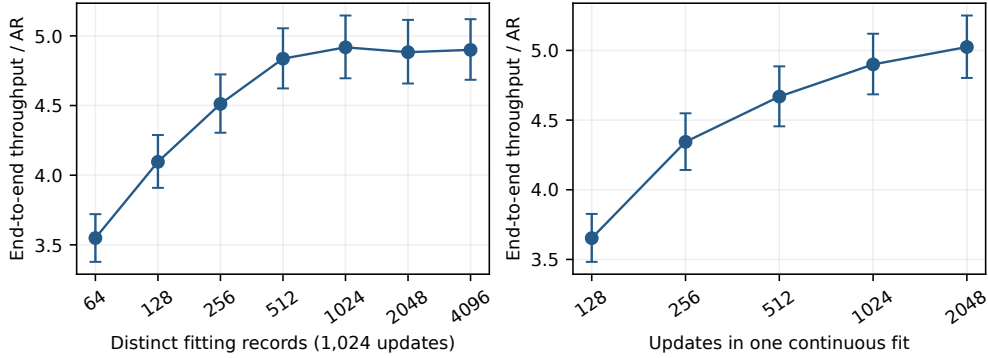


Figure 7: Historical MATH-only fitting study. Left: distinct-data scaling at 1,024 updates. Right: one continuous optimization trajectory. This uses a different training dataset and budget from Figure 4.

**Compression should account for the feature distribution.** At matched rank 1,024, approximating the mapper’s weight matrix alone can be much weaker than preserving its observed projection outputs (Table 15). For the latter, we sample up to eight feature positions from each of 512 fitting records, approximate the dominant uncentered output subspace  $U_k$  of  $XW^\top$ , and deploy  $U_k(U_k^\top W)$ , using normalized inputs in  $X$  where the mapper requires them. This is an application of data-aware compression, with established precedents in ASVD and SVD-LLM (Yuan et al., 2023; Wang et al., 2025b), not a new general factorization method. EAGLE-3 illustrates the distinction: weight-only SVD retains 54.1% on GSM8K, output-subspace compression 93.9%, and direct teacher fitting 96.1%. On MATH the latter two are nearly tied. Direct fitting remains useful alongside posthoc compression.

| Drafter / verifier | Task  | Weight SVD | Output-subspace | Direct fitting |
|--------------------|-------|------------|-----------------|----------------|
| EAGLE-3 8B         | GSM8K | 54.1%      | 93.9%           | 96.1%          |
| EAGLE-3 8B         | MATH  | 59.4%      | 96.7%           | 96.5%          |
| DFlash 14B         | MATH  | 92.7%      | 96.6%           | 97.3%          |

Table 15: Rank-1,024 throughput retained relative to the paired full map on 32 exposed questions with a 128-token cap. Posthoc arms inherit a trained dense map, while direct fitting starts from the original teacher pairs. The deployment rank is matched, not total adaptation compute.

**Retargeting toward a smaller verifier.** As a boundary experiment, we reuse the Qwen3-4B DFlash drafter with Qwen3-0.6B, taking target layers [1,7,13,19,25]. The source and target tokenizer files are byte-identical at the pinned revisions. We extract 512 fitting and 128 validation records and fit for 8,192 updates. Two dense seeds use 13.11M mapper parameters. Factored linear and GELU MLP maps at width 1,024 each use 7.86M. Per-worker fitting takes 54–61 seconds after cache extraction, which takes 32.8 seconds on four GPUs including model loading.

On 16 development GSM8K requests, dense RelaySpec reaches 168.0 tokens/s, or 3.26 times AR [3.00,3.50], compared with 102.6 tokens/s for source reuse (Table 19). The second seed repeats the result. AR time divided by RelaySpec time is 3.32 [3.07,3.55]. We report both ratios because outputs differ from AR on nine questions in this BF16 run. All methods finish below the 512-token cap. Correctness is 10/16 for speculative methods and 9/16 for AR, with a conservative paired accuracy-difference interval of  $[-23.9, 33.8]$  percentage points for dense RelaySpec. This does not establish accuracy superiority or noninferiority.

Source verification occupies 46.8% of source-reuse request time. RelaySpec removes it while reducing progress per cycle from 5.45 to 4.61. This supports the source-removal mechanism even when the verifier is smaller than the drafter’s source model. It remains a small, single-workload development result rather than a confirmed deployment recommendation for small verifiers.

To investigate the output disagreement, we repeat the same sixteen questions with the same fitted checkpoint in BF16 and full FP32. All source, draft, target and mapper computations change precision together, and FP32 disables TF32. Within-precision AR agreement is 7/16 in BF16 and 16/16 in

FP32. The BF16 and FP32 AR outputs themselves match on only 5/16 questions. This intervention supports a numerical explanation in this sample, but also changes kernel dispatch and does not isolate target precision alone. It is not a proof of agreement on other prompts. The four diagnostic workers complete in 61–78 seconds. Numerical sensitivity and FP32 mitigation have prior study, including LayerCast (Yuan et al., 2025). Our contribution here is a controlled diagnostic of the inherited-drafter interface, not discovery of this numerical phenomenon or a new precision algorithm.

The pattern extends to the tested code and dialogue screens (Table 16). Code agreement rises from 5/8 to 8/8, and dialogue agreement from 8/16 to 16/16 timed turns across eight conversations. Each precision has its own AR baseline. Code outputs are not execution-scored, and dialogue responses have no quality scores. The results support applicability beyond math in this small verifier, while distinguishing token agreement from task performance.

| Workload | Precision | Units | AR matches | Throughput / AR (95% CI) |
|----------|-----------|-------|------------|--------------------------|
| GSM8K    | BF16      | 16    | 7/16       | 3.29 [3.02, 3.54]        |
| GSM8K    | FP32      | 16    | 16/16      | 3.37 [3.12, 3.63]        |
| Code     | BF16      | 8     | 5/8        | 2.52 [2.25, 2.79]        |
| Code     | FP32      | 8     | 8/8        | 2.79 [2.50, 3.10]        |
| Dialogue | BF16      | 8     | 8/16       | 1.94 [1.58, 2.71]        |
| Dialogue | FP32      | 8     | 16/16      | 1.95 [1.57, 2.78]        |

Table 16: Qwen3-0.6B precision diagnostics with the frozen dense mapper. Units are questions or whole conversations. AR matches count timed turns. Paired intervals resample the units within each precision. GSM8K and code use 512-token caps, dialogue uses 256 tokens per turn.

We also run the same sixteen GSM8K questions with the main Qwen3-8B verifier and its frozen dense mapper. Exact AR agreement rises from 12/16 in BF16 to 16/16 in full FP32. No answer reaches the 512-token cap. The BF16 workers finish in 87–104 seconds and FP32 workers in 152–179 seconds. Runtime records confirm the requested parameter precision and disabled TF32 for FP32. Together with the smaller-verifier screens, this supports the numerical diagnosis across the two tested verifier sizes, without extending the agreement claim beyond these requests and execution settings.

**Is output-head precision sufficient?** We next promote only the verifier’s output projection to FP32, leaving the transformer, input embeddings and mapper in BF16. We copy the output head before promotion to preserve input-embedding precision if the weights are tied. On the same sixteen GSM8K questions, agreement with each runtime’s own AR rises from 7/16 to 9/16 for Qwen3-0.6B and from 12/16 to 16/16 for Qwen3-8B (Table 17). Head-only FP32 does not reproduce full-FP32 AR: those matches are 7/16 and 13/16, respectively. Thus the output projection is a useful intervention in the tested 8B setting, but does not explain all precision sensitivity or provide full-FP32 equivalence. These sixteen requests were already exposed during diagnosis.

| Target | Head | Own AR matches | FP32 AR matches | Tokens/s | Throughput / AR |
|--------|------|----------------|-----------------|----------|-----------------|
| 0.6B   | BF16 | 7/16           | 5/16            | 168.1    | 3.24            |
| 0.6B   | FP32 | 9/16           | 7/16            | 161.5    | 3.23            |
| 8B     | BF16 | 12/16          | 9/16            | 164.6    | 4.30            |
| 8B     | FP32 | 16/16          | 13/16           | 154.8    | 4.28            |

Table 17: Output-head precision with a BF16 transformer. Own AR uses the same runtime as RelaySpec. FP32 AR runs the entire model in FP32 and is a distinct reference. Rates concern the same questions but can time different outputs across precision settings.

The initial 8B agreement does not generalize to a broader development screen (Table 18). On 32 additional GSM8K requests, FP32-head agreement with own-runtime AR rises from 15/32 to 20/32. On sixteen MATH requests it rises from 3/16 to 5/16. Released native DFlash and RelaySpec have identical output strings on every request within each precision setting, including their disagreements with AR. This points to a shared execution issue in these samples, rather than an effect unique to retargeting. Head promotion is only a partial mitigation. Native DFlash also uses the target output

head for proposal logits, so its intervention changes that projection too. The MATH screen is heavily truncated at 512 tokens and does not measure full-answer accuracy.

| Workload | Head | Units | Native matches | Relay matches | AR capped |
|----------|------|-------|----------------|---------------|-----------|
| GSM8K    | BF16 | 32    | 15/32          | 15/32         | 1/32      |
| GSM8K    | FP32 | 32    | 20/32          | 20/32         | 1/32      |
| MATH     | BF16 | 16    | 3/16           | 3/16          | 11/16     |
| MATH     | FP32 | 16    | 5/16           | 5/16          | 11/16     |

Table 18: Broader 8B head-precision diagnostic. Matches compare complete recorded token sequences with each runtime’s own AR, including capped sequences. MATH AR reaches the cap on 11/16 requests in both settings. Native and RelaySpec reach it on 11/16 in BF16 and 10/16 with the FP32 head. One GSM8K request is capped for every method in both settings.

| Method               | Tokens/s | Throughput / AR | AR time / time | Correct | AR matches |
|----------------------|----------|-----------------|----------------|---------|------------|
| AR                   | 51.6     | 1.00            | 1.00           | 9/16    | 16/16      |
| Source reuse         | 102.6    | 1.99            | 2.03           | 10/16   | 7/16       |
| Dense, seed 1729     | 168.0    | 3.26            | 3.32           | 10/16   | 7/16       |
| Dense, duplicate     | 167.0    | 3.24            | 3.30           | 10/16   | 7/16       |
| Dense, seed 1730     | 168.2    | 3.26            | 3.32           | 10/16   | 7/16       |
| Factored linear-1024 | 161.2    | 3.13            | 3.18           | 10/16   | 7/16       |
| MLP-1024             | 147.9    | 2.87            | 2.92           | 10/16   | 7/16       |

Table 19: Downward retargeting to Qwen3-0.6B on 16 GSM8K requests. Timing includes prompt processing. Correctness and exact AR-output matches are separate. The duplicate row repeats the same checkpoint and passes the trajectory-equivalence check. All models run in BF16.

**Selecting target features.** The interface also permits a controlled test of target-layer sufficiency. We refit DFlash’s linear mapper using only target layers 25 and 33, keeping the released drafter frozen, the same 512 fitting records and 8,192 batch-four updates. This reduces the mapper from 52.43M to 20.97M parameters. It does not remove target-model layers or reduce whole-model parameters by 60%. Initial development screens selected this two-layer interface before confirmation.

A separate protocol then fixed both checkpoint hashes, the first 64 lexicographically ordered IDs from the previously unused GSM8K reserve, a 2,048-token cap, and a 95% throughput-retention criterion. An updated exposure audit found no reserve IDs in 964 collected rank files across both worktrees. The original fitting-data exact/lexical exclusions remain inherited. Four disjoint 16-question shards each compare the same two methods on one L40S. The paired bootstrap resamples all 64 requests 10,000 times with seed 1729. The lower 95% endpoint exceeds the fixed 95% retention threshold (Table 20). Both maps produce identical token sequences on all 64 questions and score 62/64. None reaches the cap. A conservative paired accuracy-difference interval is  $[-6.62, 6.62]$  percentage points, so this sample does not establish one-point quality noninferiority.

| Interface          | Parameters | Tokens/s | Retention (95% CI)       | Correct |
|--------------------|------------|----------|--------------------------|---------|
| Five target layers | 52.43M     | 160.67   | 100.00 [100.00, 100.00]% | 62/64   |
| Two target layers  | 20.97M     | 156.86   | 97.63 [96.23, 99.03]%    | 62/64   |

Table 20: Fixed DFlash-8B interface comparison on 64 fresh GSM8K requests. Parameter counts concern the mapper. Throughput includes request overhead. Intervals resample requests and do not include fitting-seed variation.

The recorded profiles explain why reducing mapper size need not improve end-to-end speed. In this fixed comparison, full-map computation occupies 0.74% of total request time, including its prefill application, while full-target verification occupies 81.9%. The two-layer map reduces its own share to 0.58%, but modestly lower progress per verification cycle raises total verification time. Preserving

acceptance therefore matters more than mapper arithmetic in this setting. These are recorded region attributions, not isolated kernel or memory measurements.

The result has a family-dependent boundary. On 16 exposed, 128-token MATH requests, two-layer refitting retains 99.3% of full-map throughput with a 14B DFlash verifier, versus 93.7% with EAGLE-3. A second 8B fitting seed retains 97.6% for DFlash and 94.8% for EAGLE-3 on 16 exposed GSM8K requests. These are development screens, not additional confirmation tests. They motivate studying which features a frozen interface requires rather than asserting that two target layers always suffice.

At equal two-layer mapper size, depth matters (Figure 8). Refitting from early layers 1 and 9 retains 56.3% of full-map throughput for DFlash and 57.0% for EAGLE-3. Spaced layers 9 and 25 retain 91.5% and 91.4%, while late layers 25 and 33 retain 98.7% and 96.2% on the same 16 development questions. All fits use 512 records, 8,192 updates and seed 1729. This controls mapper capacity, supporting an information-depth explanation within these settings. It does not establish that this particular layer pair is optimal across tasks or models.

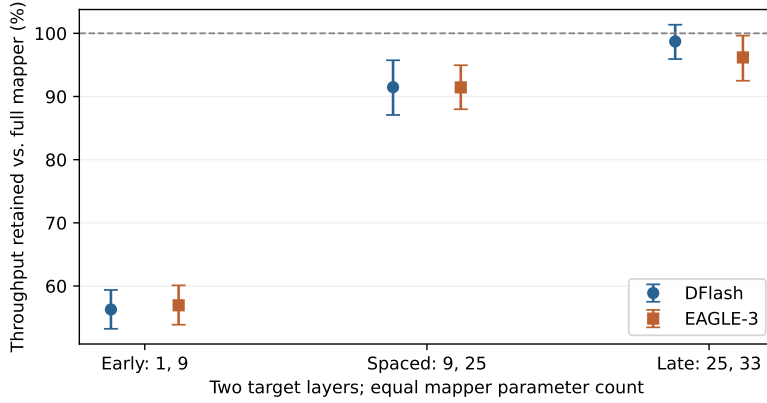


Figure 8: Equal-capacity target-layer comparison on 16 exposed GSM8K questions with a 128-token cap. Each arm is paired with its own full-map baseline. Late-arm rows are the matching subset of the earlier 32-request screen. Paired 95% request intervals exclude fitting-seed variation.

**Can nonlinearity compensate for early-layer inputs?** We fit an early-layer GELU MLP and a factorized linear map at width 1,950, giving exactly 20,966,400 parameters in each case. Both use layers [1,9], 512 records and 8,192 updates at seed 1729. Each decoding worker also evaluates the existing early- and late-layer dense maps. The MLP lowers training error in both drafter families, but does not close the gap to late-layer inputs (Table 21). It worsens DFlash validation error. For EAGLE-3 it slightly improves validation error, yet throughput remains far below the late-layer control. Late-layer dense maps retain 97.4–98.6% in the DFlash workers and 92.6–92.7% in EAGLE-3, compared with 50.0–57.9% for the fitted early-layer maps. This limits the tested nonlinear fitting approach, not the recoverability of later-layer information in principle.

| Family  | Early-layer map | Train error | Val. error | Progress/cycle | Retention |
|---------|-----------------|-------------|------------|----------------|-----------|
| DFlash  | Factored linear | 0.333       | 0.409      | 2.61           | 56.0%     |
| DFlash  | GELU MLP        | 0.271       | 0.439      | 2.33           | 50.0%     |
| EAGLE-3 | Factored linear | 0.333       | 0.411      | 2.57           | 57.9%     |
| EAGLE-3 | GELU MLP        | 0.272       | 0.399      | 2.56           | 57.4%     |

Table 21: Early-layer reconstruction at matched parameter count. Retention is relative to each worker’s paired full map on eight exposed GSM8K requests with a 128-token cap. Progress includes the committed target token. Errors use the same fitting and validation diagnostics within each family. One fitting seed is tested.

**Cross-domain limits of layer sufficiency.** Without changing the Numina-fitted maps, we repeat the depth screen on code and dialogue (Table 22). DFlash’s late-layer map retains 99.7% on code and 97.8% on dialogue. EAGLE-3 retains 85.4% and 91.9%, and its spaced-layer map is slightly faster on dialogue. Thus late-layer sufficiency transfers well in the tested DFlash screens, but is not a family-independent rule. These small exposed screens do not establish code test-pass rates, dialogue quality, or a generally optimal layer choice.

| Drafter | Workload | Early | Spaced | Late  |
|---------|----------|-------|--------|-------|
| DFlash  | Code     | 40.5% | 84.8%  | 99.7% |
| DFlash  | Dialogue | 65.8% | 91.3%  | 97.8% |
| EAGLE-3 | Code     | 52.0% | 84.9%  | 85.4% |
| EAGLE-3 | Dialogue | 69.6% | 93.0%  | 91.9% |

Table 22: Throughput retained relative to each arm’s paired full mapper. Code uses eight MBPP requests with a 512-token cap. Dialogue uses eight two-turn MT-Bench conversations with a 256-token per-turn cap. Early, spaced and late denote layers [1, 9], [9, 25] and [25, 33]. All compact maps have equal parameter count and the same fitting data and budget.

**Native interface compression as an application.** The same interface abstraction supports a secondary experiment without retargeting: compress the released Qwen3-8B DFlash input projection by rank-1,536 SVD, leaving the rest of the native drafter unchanged. The rank was fixed from development screens before generating any of the 64 confirmation answers above. We retain both the released native path and a full-rank projection repacked into the relay path as controls. The compressed projection has 37.75M rather than 83.89M parameters (55% fewer). This is not a whole-drafter parameter reduction.

Table 23 uses the same 64 requests and output cap. Relative to the released native path, compression retains 98.66% [97.58, 99.75]% end-to-end throughput. All three methods score 62/64 with identical outputs and no cap-length outputs. Repacking alone retains 99.16%, so the table distinguishes packaging overhead from compression. This is a preselected secondary comparison, not another primary significance test. It demonstrates an application of inspecting and replacing the feature interface. SVD itself is not a novel method, and activation-aware LLM compression already has established precedents (Yuan et al., 2023; Wang et al., 2025b).

| Interface       | Parameters | Tokens/s | Retention (95% CI)       | Correct |
|-----------------|------------|----------|--------------------------|---------|
| Released native | 83.89M     | 178.65   | 100.00 [100.00, 100.00]% | 62/64   |
| Repacked native | 83.89M     | 177.16   | 99.16 [99.00, 99.28]%    | 62/64   |
| Rank 1,536      | 37.75M     | 176.25   | 98.66 [97.58, 99.75]%    | 62/64   |

Table 23: Preselected native DFlash projection compression on 64 GSM8K requests. Retention is relative to the released native implementation. Only projection parameters are counted.

**Distilling a native interface from fewer layers.** The layer result also suggests an alternative to weight decomposition: fit a smaller interface directly to the released native projection’s outputs. Using cached target features from 512 fitting records, we apply the frozen native projection and its output normalization to construct supervision, then fit a one-layer or two-layer input map for 8,192 updates. The native drafter remains frozen. Teacher hashes, target identities and layer lists are checked, and the fitting path passes its batched versus individual-example gradient check.

The two-layer native map has 33.55M parameters (60% fewer than the 83.89M native projection), while one layer has 16.78M (80% fewer). Table 24 shows the tradeoff on development workloads: two layers retain 97.1–97.9% throughput relative to released native, while one layer loses substantially more, especially on code. Rank-1,536 SVD retains slightly more throughput with slightly more parameters (37.75M). Thus the student provides another compact-interface operating point, not a uniform improvement over decomposition. These students were selected after the fixed native-SVD comparison and are not evaluated as new candidates on its confirmation questions.

| Workload | Units | One layer | Two layers | SVD rank 1,536 |
|----------|-------|-----------|------------|----------------|
| GSM8K    | 32    | 84.7%     | 97.7%      | 98.8%          |
| MATH     | 32    | 88.8%     | 97.1%      | 98.1%          |
| Code     | 8     | 71.6%     | 97.2%      | 98.6%          |
| Dialogue | 8     | 87.0%     | 97.9%      | 99.2%          |

Table 24: Native interface students versus released-native throughput. GSM8K/MATH use 32 questions and a 128-token cap, code eight MBPP requests and a 512-token cap, and dialogue eight two-turn conversations with a 256-token per-turn cap. All are exposed development workloads.

**Executable-code development check.** A separate fixed set of 32 MBPP tasks compares all five frozen native variants at a 1,024-token cap. These tasks exclude the earlier autoresearch MBPP IDs but appear in historical evaluations, so they are development examples. Every variant produces the same tokens and extracted code, passes 28/32 published base-test suites and has no cap-length outputs. The four failures are shared. Tests execute in an isolated Python process using the first Python code block, without EvalPlus extended tests. Two layers retain 95.4% [93.5,97.3]% of released-native throughput, versus 71.6% for one layer and 97.7% for rank-1,536 SVD. This supports observed functional agreement alongside a workload-dependent efficiency penalty. The conservative paired accuracy interval is  $[-12.80, +12.80]$  percentage points, not tight quality noninferiority.

**Calibration data depends on the interface role.** We fit native and retargeted two-layer maps on the same 16- or 128-record prefixes, at 8,192 batch-four updates and two fitting seeds. Each map is compared with its role’s original 512-record two-layer reference on the same eight exposed GSM8K questions, with a 512-token cap. At 128 records, native maps retain 97.2–97.9% throughput across seeds, versus 87.7–88.9% for retargeting. At 16 records these ranges fall to 66.7–67.0% and 46.3–47.1%, respectively. The 16-record fits closely match training features but have much larger validation errors. These standard recipes retain different teacher outputs, output widths and normalization, so the contrast does not isolate an intrinsic difference in sample complexity. The two seeds share questions and reference maps. This is a calibration-sensitivity screen, not a minimum-data guarantee or independent quality confirmation.

**Inherited weights change the calibration requirement.** A native interface permits an additional control: retain the released projection columns for layers 25 and 33, discarding the other columns. This untrained crop already preserves much of native throughput (Table 25). Initializing the same compact map from these columns instead of random weights improves sparse-data fitting. At 128 updates on 16 records, validation error falls from 0.347 to 0.038 and GSM8K throughput retention rises from 51.7% to 99.3%, relative to the original 512-record compact map. At 128 records the corresponding retentions are 72.4% and 101.0%. All non-initialization fitting settings are matched. The two 128-update inherited fits take 1.55–1.58 seconds of optimizer-update time and 8.8–10.4 seconds within the fitting worker, excluding frozen-feature extraction and original drafter training. Longer inherited fits at 1,024 and 8,192 updates give no clear GSM8K throughput advantage in this screen. Table 25 pairs the short and long fit candidates with untrained cropping across four workloads. This separates inherited performance from the additional effect of calibration. All paired outputs match, but MATH and dialogue include capped outputs, and code and dialogue quality are unscored. Each workload has eight exposed requests, with two turns per dialogue conversation. The fixed record order and copied initialization produce identical weight storage when the nominal fitting seed changes, so those repeats are not independent fitting-seed evidence. Four additional disjoint 16-record training windows, verified against the frozen cache index, retain 99.1–100.0% of the 512-record compact reference after 128 updates on the same GSM8K screen. This checks calibration-record sensitivity without adding independent test questions. This is an initialization-dependent calibration finding, not a universal minimum-data claim or a new general pruning algorithm.

| Task     | Updates | Crop | Rnd16 | Inh16 | Rnd128 | Inh128 | Ref512 |
|----------|---------|------|-------|-------|--------|--------|--------|
| GSM8K    | 8,192   | 95.7 | 66.6  | 99.8  | 96.7   | 100.6  | 99.8   |
| MATH     | 8,192   | 90.8 | 69.0  | 93.5  | 95.5   | 96.2   | 97.4   |
| Code     | 8,192   | 90.6 | 52.6  | 92.6  | 92.6   | 94.2   | 98.0   |
| Dialogue | 8,192   | 98.4 | 72.7  | 98.4  | 96.2   | 98.1   | 98.8   |
| GSM8K    | 128     | 95.9 | 51.6  | 98.6  | 72.4   | 100.6  | 99.9   |
| MATH     | 128     | 90.8 | 52.0  | 94.5  | 76.2   | 95.2   | 97.4   |
| Code     | 128     | 90.8 | 36.8  | 94.4  | 56.2   | 95.2   | 98.2   |
| Dialogue | 128     | 98.4 | 63.1  | 97.7  | 75.0   | 99.2   | 98.8   |

Table 25: Native initialization controls. Values are throughput percentages relative to each worker’s full repacked-native projection. Crop uses no fitting. Rnd and Inh denote random and inherited-column initialization, with 16 or 128 records. Ref512 is the original 512-record compact map. All compact maps retain two layers and 33.55M projection weights instead of 83.89M. Each row pairs all seven methods. Short and long budgets are separate campaigns on the same exposed requests, with a 512-token cap. Full paired request intervals are retained in the accompanying artifacts.

**A scalar-reweighting control.** We freeze the cropped native columns and learn only one gain per retained layer, folding both gains into the exported projection. All four settings (16/128 records, 128 updates, learning rates 0.003/0.03) are paired with untrained cropping and full-matrix fits on the same four development workloads. Scalar/crop throughput ratios range from 99.8% to 100.8%. All 16 paired 95% intervals include one. On MATH, scalar fits retain 95.7–96.6% of full-matrix fitting throughput at the same record count, with all four paired intervals below one. These exposed, capped screens suggest that the fitted feature transformation adds more than layer reweighting in this setting. They do not establish superiority over all scalar optimizers: full fits use learning rate 0.0006, and intervals exclude selection and multiple-comparison uncertainty.

**A separately frozen short-fit test.** Before decoding, we fix the 16-record, 128-update inherited map as the primary candidate on 64 further GSM8K questions (sorted reserve positions 128–191). An exact-ID audit finds no matches in 2,726 prior local result files. This is not a global or pretraining exposure guarantee. The predeclared criterion requires the lower paired 95% throughput-retention bound against released native to reach 95%. The primary candidate retains 95.09% [94.03,96.13]% and *fails* this criterion (Table 26). The 128-record control retains 96.60% [95.48,97.70]%, but remains descriptive rather than replacing the primary candidate. All five methods produce identical outputs, score 62/64, and avoid the 2,048-token cap. The conservative paired accuracy interval remains [−6.62, +6.62] percentage points. No optional extension or refitting uses these outcomes. Thus the short fits support efficient calibration, while the fresh test limits a 16-record sufficiency claim inferred from the smaller development screen.

| Interface                | Parameters | Tokens/s | Retention (95% CI)       | Correct |
|--------------------------|------------|----------|--------------------------|---------|
| Released native          | 83.89M     | 158.99   | 100.00 [100.00, 100.00]% | 62/64   |
| Repacked native          | 83.89M     | 157.66   | 99.16 [99.08, 99.24]%    | 62/64   |
| Untrained crop           | 33.55M     | 147.50   | 92.77 [91.62, 93.93]%    | 62/64   |
| 16 records, 128 updates  | 33.55M     | 151.19   | 95.09 [94.03, 96.13]%    | 62/64   |
| 128 records, 128 updates | 33.55M     | 153.59   | 96.60 [95.48, 97.70]%    | 62/64   |

Table 26: Frozen short-fit comparison on 64 additional GSM8K questions. Only the 16-record map is primary. All fitted compact maps inherit native columns and use 128 updates. Intervals use 10,000 paired question bootstrap replicates, seed 1729, and exclude fitting-data and fitting-seed variation.

**Frozen native-student confirmation.** We freeze the two-layer student’s checkpoint and a new 64-question set before decoding. These are sorted reserve positions 64–127, disjoint from the preceding interface/SVD confirmation. A scan of 932 prior local result files finds no matching question IDs. This does not establish global or pretraining non-exposure. The primary criterion requires the lower endpoint of the paired 95% throughput interval against released native to exceed 95%. We retain the one-layer student, rank-1,536 SVD and full repacked projection as descriptive controls, with no optional sample extension or candidate refitting.



The two-layer student retains 97.05% [95.93,98.21]% of released-native throughput and passes the fixed criterion (Table 27). One layer retains only 84.55% [83.11,86.10]%, while SVD retains 98.08% [97.13,99.06]% with slightly more parameters than two layers. Every method produces identical recorded outputs, scoring 60/64, and none reaches the 2,048-token cap. The conservative paired accuracy interval for two layers versus released native is still  $[-6.62, +6.62]$  percentage points: identical observed answers do not establish tight accuracy noninferiority. Together with retargeting, this supports a compact late-layer conditioning interface in two uses, without claiming that two layers suffice for every drafter or verifier.

| Interface         | Parameters | Tokens/s | Retention (95% CI)       | Correct |
|-------------------|------------|----------|--------------------------|---------|
| Released native   | 83.89M     | 170.22   | 100.00 [100.00, 100.00]% | 60/64   |
| Repacked native   | 83.89M     | 168.91   | 99.23 [99.18, 99.28]%    | 60/64   |
| Two-layer student | 33.55M     | 165.21   | 97.05 [95.93, 98.21]%    | 60/64   |
| One-layer student | 16.78M     | 143.93   | 84.55 [83.11, 86.10]%    | 60/64   |
| Rank 1,536        | 37.75M     | 166.96   | 98.08 [97.13, 99.06]%    | 60/64   |

Table 27: Fixed native-student comparison on a separate set of 64 GSM8K questions. Retention uses the released-native reference and 10,000 paired question bootstrap replicates, seed 1729. Only the two-layer comparison is primary. Counts refer to the input projection, not the whole drafter. Fitting-seed variation is outside these intervals.

**A smaller-verifier boundary.** Repeating the depth screen with the 0.6B verifier gives a weaker compact interface (Table 28). At matched 5.24M parameters, early, spaced and late two-layer maps retain 52.5%, 79.9% and 82.2% of the full five-layer mapper’s throughput. One layer retains 60.7% with 2.62M parameters. Late features remain more useful in this screen, but the near-full retention observed at 8B does not carry over to this smaller verifier. This is an eight-question development result with one fitting seed, not a general lower bound on the required layers.

| Verifier   | Target layers | Train error | Val. error | Progress/cycle | Retention |
|------------|---------------|-------------|------------|----------------|-----------|
| Qwen3-0.6B | Early [1,7]   | 0.432       | 0.459      | 2.34           | 52.5%     |
| Qwen3-0.6B | Spaced [7,19] | 0.355       | 0.374      | 3.70           | 79.9%     |
| Qwen3-0.6B | Late [19,25]  | 0.332       | 0.351      | 3.76           | 82.2%     |
| Qwen3-0.6B | Last [25]     | 0.417       | 0.430      | 2.75           | 60.7%     |

Table 28: Downward-retargeting depth screen, 4B-trained DFlash to a 0.6B verifier. All fits use 512 records and 8,192 updates. Decoding uses eight exposed GSM8K requests with a 512-token cap. Retention is relative to each worker’s paired full mapper. Errors use the common fitting and validation sets.

A matched-capacity nonlinear follow-up does not recover that deficit (Table 29). Width-1,137 MLP and factorized maps each have 5,239,296 parameters, versus 5,242,880 for late dense. MLPs lower training error from about 0.332 to 0.303–0.304 and validation error from 0.351 to 0.345, yet retain only 81.1–83.1% throughput at two fitting seeds. Factorized maps retain 83.1–84.2%. Their paired late-dense controls retain 82.8–84.3%. The same eight questions are reused across seeds. This closes the tested fitting intervention, without establishing that later features are unrecoverable in principle.

| Seed | Architecture      | Train error | Val. error | Progress/cycle | Retention |
|------|-------------------|-------------|------------|----------------|-----------|
| 1729 | GELU MLP          | 0.304       | 0.345      | 3.69           | 81.1%     |
| 1729 | Factorized linear | 0.336       | 0.353      | 3.76           | 84.2%     |
| 1730 | GELU MLP          | 0.303       | 0.345      | 3.78           | 83.1%     |
| 1730 | Factorized linear | 0.336       | 0.353      | 3.77           | 83.1%     |

Table 29: Matched-size nonlinear follow-up for 0.6B late-layer mapping. Both architectures use layers [19,25], 512 records and 8,192 updates. Retention is relative to each worker’s full mapper on the same eight exposed questions. Fitting seeds do not add independent evaluation samples. Better feature regression does not close the decoding deficit.

**Expanded NuminaMath data curve.** Table 30 reports all eleven completed sizes at 8,192 batch-four updates, spanning 2,048 epochs at 16 records to one epoch at 32,768. The same 1,024 feature-validation records are used throughout. Larger data marginally improves feature loss but does not improve measured decoding beyond the plateau. The 512-record mapper retains 97.2% of the observed best throughput, while the 256-record mapper retains 93.9%. These percentages are descriptive and conditional on the evaluated fitting seed and development questions.

Table 30: NuminaMath fitting at fixed 8,192 updates on 128 paired MATH questions with a 2,048-token output cap. All eleven mapper outputs score 105/128, versus AR’s 104/128.

| Records | Tokens/s | Speedup / AR | Correct / 128 |
|---------|----------|--------------|---------------|
| 16      | 96.87    | 2.58×        | 105           |
| 32      | 133.23   | 3.55×        | 105           |
| 64      | 157.74   | 4.20×        | 105           |
| 128     | 177.75   | 4.73×        | 105           |
| 256     | 186.36   | 4.96×        | 105           |
| 512     | 192.81   | 5.14×        | 105           |
| 2,048   | 196.93   | 5.25×        | 105           |
| 4,096   | 198.30   | 5.28×        | 105           |
| 8,192   | 197.98   | 5.27×        | 105           |
| 16,384  | 198.37   | 5.28×        | 105           |
| 32,768  | 197.94   | 5.27×        | 105           |

Table 31: Completed DFlash-8B fitting studies on the same 128-question set. Records count distinct fitting examples. Two passes over 4,096 records give 8,192 presentations. Source reuse is paired within each run.

| Fitting choice            | Records | Updates      | Relay tok/s | Relay / source [95% CI] | Score |
|---------------------------|---------|--------------|-------------|-------------------------|-------|
| 512 records               | 512     | 128          | 136.78      | 1.150 [1.118, 1.181]    | 82.03 |
| 1,024 records             | 1,024   | 256          | 162.04      | 1.372 [1.347, 1.397]    | 82.03 |
| 2,048 records             | 2,048   | 512          | 173.72      | 1.477 [1.458, 1.495]    | 82.03 |
| 4,096 records, two passes | 4,096   | 2,048        | 187.85      | 1.573 [1.558, 1.588]    | 82.03 |
| Ridge surrogate           | 4,096   | <i>solve</i> | 115.18      | 0.955 [0.927, 0.982]    | 82.03 |
| Nonlinear width 512       | 4,096   | 1,024        | 104.49      | 0.872 [0.841, 0.903]    | 82.03 |

The data-budget configurations use the same normalized linear interface and relative-error objective. The nonlinear map uses an intermediate width of 512 and about 11.8M parameters, compared with 52.4M for the dense 8B map. Its lower throughput is consistent with a less effective fitted interface, but the comparison also changes capacity. It does not isolate a disadvantage of nonlinear functions. The ridge configuration solves a raw linear surrogate with ridge coefficient 1.0. Since the deployed DFlash interface includes output normalization, this is a different objective from Equation 3.

**Matched capacity on a separate fitting source.** We additionally fit 30 DFlash-8B configurations on nested sets of 512 and 2,048 NuminaMath-CoT records,<sup>1</sup> using a fixed 1,024-record fitting validation split. The source-stratified sets include `cn_k12`, `orca_math` and `synthetic_math` examples. Their lexical filter excludes exact normalized problem/solution matches and near matches at five-shingle Jaccard similarity at least 0.6 for texts with at least ten unique shingles. This filter does not establish semantic or template independence. These results use a separate fitting source from the historical MATH study above.

Each configuration receives 8,192 updates with four records per update: 64 passes at  $N = 512$  and 16 at  $N = 2048$ . Frozen features are cached once. Factored linear maps and bias-free two-layer GELU MLPs use widths 64, 128, 256, 512, 1,024, 2,048 and 4,096, giving matched parameter counts  $h(20,480 + 2,560)$ . The dense reference has 52.43M parameters. The rank of a linear function remains bounded by 2,560 even when its factorization width is larger. All cells share the data order, relative-interface objective, learning rate  $6 \times 10^{-4}$  and seed 1729.

<sup>1</sup>AI-MO/NuminaMath-CoT, revision 9d8d210c9f6a, training shard 0 of 5.

Table 32: Validation relative interface MSE for the capacity study. Linear and MLP entries at the same width have equal parameter counts. The dense row supplies the single-linear-layer reference.

| Width | Params (M) | Linear, $N=512$ | MLP, $N=512$ | Linear, $N=2048$ | MLP, $N=2048$ |
|-------|------------|-----------------|--------------|------------------|---------------|
| 64    | 1.47       | 0.5732          | 0.5790       | 0.5692           | 0.5752        |
| 128   | 2.95       | 0.4761          | 0.4871       | 0.4690           | 0.4804        |
| 256   | 5.90       | 0.3760          | 0.3883       | 0.3648           | 0.3767        |
| 512   | 11.80      | 0.2862          | 0.3082       | 0.2685           | 0.2878        |
| 1024  | 23.59      | 0.2295          | 0.2579       | 0.2040           | 0.2224        |
| 2048  | 47.19      | 0.2133          | 0.2438       | 0.1847           | 0.1982        |
| 4096  | 94.37      | 0.2145          | 0.2501       | 0.1845           | 0.1936        |
| Dense | 52.43      | 0.2101          | —            | 0.1805           | —             |

At this optimization budget, wider factorizations approach the dense reference, while MLPs have higher validation error at every matched width. Increasing MLP width from 2,048 to 4,096 at  $N = 512$  lowers diagnostic training error from 0.1586 to 0.1278 but raises validation error from 0.2438 to 0.2501. At  $N = 2048$ , validation error instead decreases from 0.1982 to 0.1936. This interaction motivates checking regularization and convergence before attributing the ranking to function class.

**More epochs on fixed smaller datasets.** We continue the width-512 linear and MLP fits to 32,768 total updates, preserving Adam moments, random states and cyclic data position. A separate bounded pilot reproduced uninterrupted fitting bit for bit. Figure 9 shows late-training diagnostics on the same first 256 training records and all 1,024 validation records. The  $N = 2048$  MLP improves from 0.2878 to 0.2787 validation error between 8,192 and 32,768 updates. At  $N = 512$ , linear training error falls from 0.2443 to 0.2350 while validation error rises from 0.2862 to 0.2873. The latter is consistent with mild overfitting on this diagnostic split. These single-seed fitting trajectories require separate decoding and quality evaluation, including replication across targets and workloads.

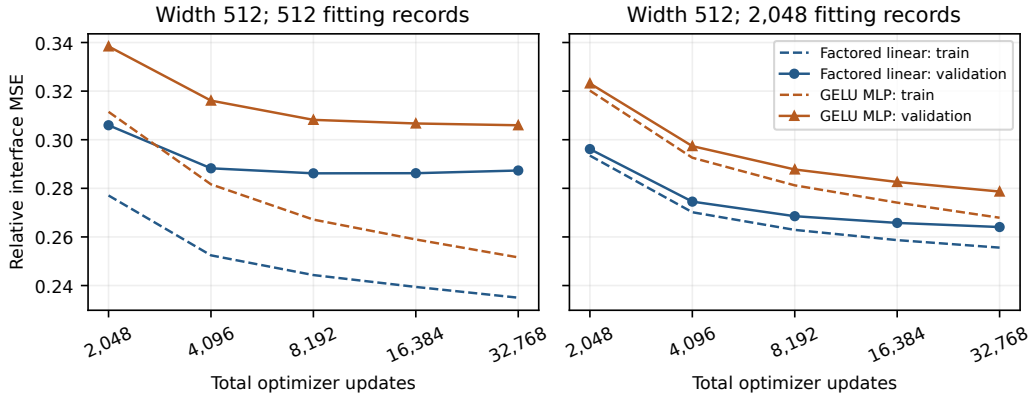


Figure 9: Late-training trajectories, starting at 2,048 updates. Dashed lines use the fixed training diagnostic subset. Solid lines use fitting validation. The horizontal axis counts updates, while the distinct datasets remain fixed at 512 or 2,048 records.

**Explicit regularization on fixed smaller datasets.** We complete 42 nonzero-penalty fits for the dense, width-512 factored linear and width-512 GELU MLP connectors at  $N \in \{512, 2048\}$ . Each fit uses 8,192 updates, the same 1,024 validation records and seed 1729. Explicit L2 coefficients are  $10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}$ . Separate AdamW arms use decay  $10^{-4}, 10^{-3}, 10^{-2}$ . No tested L2 coefficient improves its matched zero-penalty endpoint. The largest AdamW improvement is a 0.17% relative decrease in validation error for the width-512 linear connector at  $N = 512$ . Neither MLP setting improves with either penalty family. Table 33 reports the best validation value within each penalty family, so these minima are development selections. Figure 10 retains all tested coefficients.

This grid does not establish a decoding gain or a penalty effect across seeds, and it does not resolve regularization for the wider MLPs.

Table 33: Matched regularization endpoints. Lower relative interface MSE is better. Each nonzero minimum is selected from four L2 or three AdamW coefficients using the same fitting validation split.

| $N$  | Mapper              | No penalty | Best tested L2 | Best tested AdamW |
|------|---------------------|------------|----------------|-------------------|
| 512  | Dense               | 0.21008    | 0.21138        | 0.21011           |
| 512  | Factored linear 512 | 0.28617    | 0.28738        | 0.28570           |
| 512  | GELU MLP 512        | 0.30818    | 0.30827        | 0.30822           |
| 2048 | Dense               | 0.18055    | 0.18424        | 0.18055           |
| 2048 | Factored linear 512 | 0.26853    | 0.26970        | 0.26852           |
| 2048 | GELU MLP 512        | 0.28776    | 0.28864        | 0.28779           |

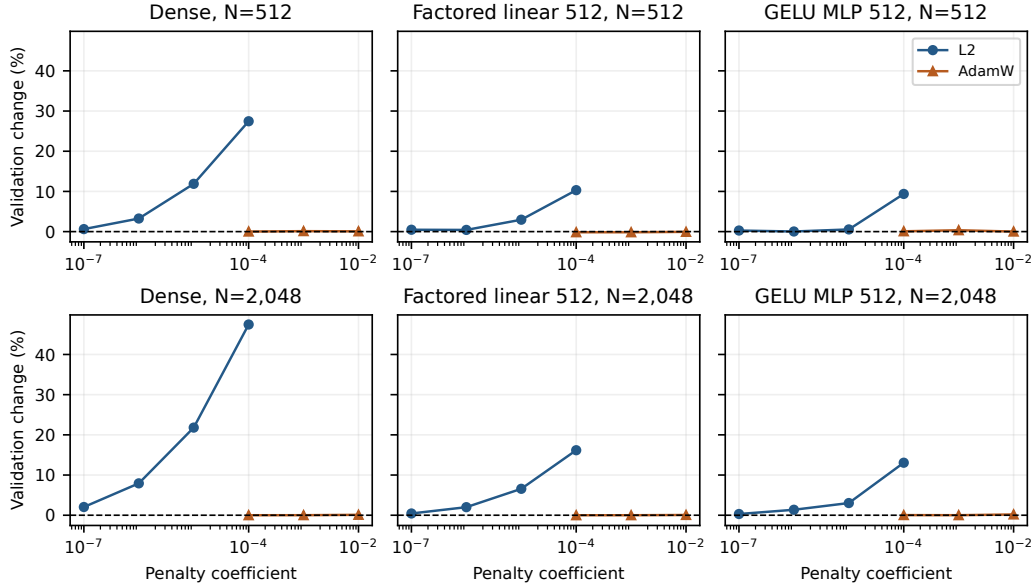


Figure 10: All 42 penalty endpoints relative to their matched zero-penalty validation error. Positive values indicate worse fitting validation. The two curves are separate objective-L2 and AdamW experiments.

Table 34: Matched EAGLE-3 8B normalization comparison on 32 development questions after 128 updates. Throughput ratios use source reuse.

| Input to linear map | Relative error | Throughput ratio [95% interval] | Progress retained |
|---------------------|----------------|---------------------------------|-------------------|
| Normalized features | 0.413          | 1.035 [1.003, 1.068]            | 69.0%             |
| Raw features        | 0.266          | 1.237 [1.213, 1.263]            | 82.6%             |

Table 35: DFlash objective comparison on 32 development questions. Candidates share fitting data, update count, normalization and optimizer. Each ratio uses the source-reuse arm of that candidate’s paired run.

| Target | Objective              | Throughput ratio [95% interval] | Token matches |
|--------|------------------------|---------------------------------|---------------|
| 8B     | Relative error         | 1.523 [1.488, 1.558]            | 32/32         |
| 8B     | Squared error + cosine | 1.509 [1.476, 1.545]            | 32/32         |
| 14B    | Relative error         | 1.265 [1.243, 1.289]            | 32/32         |
| 14B    | Squared error + cosine | 1.251 [1.229, 1.276]            | 32/32         |

Relative error weights a record’s feature mismatch by teacher magnitude. Squared error plus 0.1 times cosine distance combines coordinate error with directional agreement. Direct candidate-to-candidate throughput ratios are 1.0085 [1.0005, 1.0174] at 8B and 1.0114 [1.0046, 1.0177] at 14B. These are modest gains with one fewer mixing coefficient.

### C.1 EAGLE-3 FITTING REPLICATION

We repeat the fixed smaller-data fitting study with the EAGLE-3 source drafter and the 8B target. The grid contains dense, factored linear and GELU MLP connectors, with widths 128, 512 and 2048 for the latter two. Each primary fit uses 8,192 updates at  $N \in \{512, 2048\}$  and the same 1,024 fitting-validation records. Two additional seeds repeat the dense  $N = 2048$  setting. This experiment uses the EAGLE input convention without the DFlash input normalization. Error values are interpreted within each drafter family’s conditioning interface.

Table 36: EAGLE-3 fitting validation. End denotes the 8,192-update endpoint. Best denotes the minimum saved fitting-validation error, with its update count. These are development selections for seed 1729, not decoding results. M weights counts millions of trainable mapper parameters.

| Mapper      | M weights | $N = 512$ |         |        | $N = 2048$ |         |        |
|-------------|-----------|-----------|---------|--------|------------|---------|--------|
|             |           | End       | Best    | Update | End        | Best    | Update |
| Dense       | 52.43     | 0.24787   | 0.21666 | 1024   | 0.21203    | 0.21061 | 2048   |
| Linear 128  | 2.95      | 0.41980   | 0.41980 | 8192   | 0.41505    | 0.41505 | 8192   |
| Linear 512  | 11.80     | 0.28632   | 0.28505 | 2048   | 0.27188    | 0.27188 | 8192   |
| Linear 2048 | 47.19     | 0.23486   | 0.22877 | 2048   | 0.21132    | 0.21132 | 8192   |
| MLP 128     | 2.95      | 0.46554   | 0.46554 | 8192   | 0.45645    | 0.45645 | 8192   |
| MLP 512     | 11.80     | 0.31749   | 0.31749 | 8192   | 0.29577    | 0.29577 | 8192   |
| MLP 2048    | 47.19     | 0.28154   | 0.27124 | 2048   | 0.23196    | 0.22683 | 4096   |

The dense  $N = 512$  mapper reaches 0.21666 at 1,024 updates, then rises to 0.24787 at the final update. Its endpoint is worse than the width-2048 linear mapper, but its best saved validation value is better than that linear mapper’s 0.22877. At  $N = 2048$ , the dense and width-2048 linear minima are 0.21061 and 0.21132. Thus endpoint rankings alone can reflect different convergence trajectories. At every matched width in this grid, the MLP’s best saved validation error exceeds the linear mapper’s minimum. The three dense  $N = 2048$  endpoints range from 0.212034 to 0.212043. This narrow seed spread concerns one fitting setting. It does not establish seed stability for the other architectures or for decoding performance.

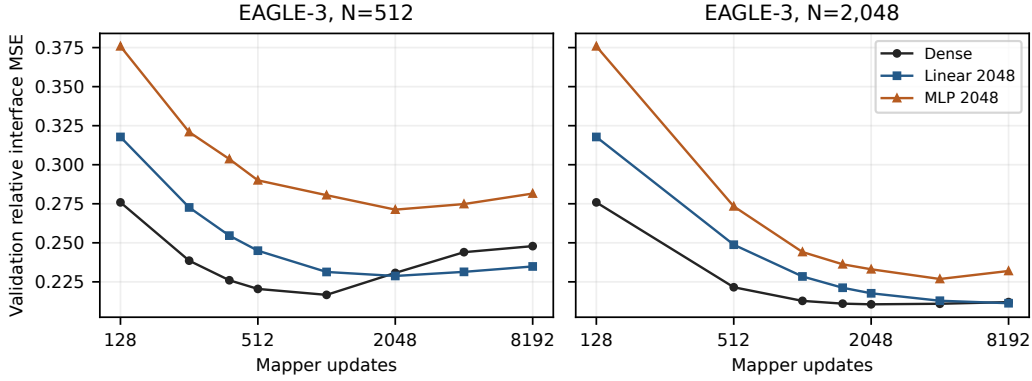


Figure 11: EAGLE-3 fitting-validation trajectories at fixed distinct-data counts. Additional updates can worsen validation, most visibly for the dense mapper at  $N = 512$ . The horizontal axis counts updates rather than new examples. The subsequent checkpoint comparison measures actual decoding.

## C.2 FEATURE-VALIDATION MINIMA AND DECODING

We compare each distinct saved feature-validation minimum with its 8,192-update endpoint in two declared decoding campaigns. Each uses 16 paired development questions, greedy decoding and a 256-token cap. Checkpoint selection precedes decoding. Table 37 includes every distinct early-versus-endpoint pair in these campaigns.

Table 37: Decoding throughput at the minimum feature-validation checkpoint relative to the same fit’s endpoint. Intervals resample paired requests 10,000 times. They are descriptive, without multiple-comparison adjustment, and do not measure variation across fitting seeds.

| Family  | Mapper      | $N$  | Learning rate | Selected update | Early / end [95% CI] |
|---------|-------------|------|---------------|-----------------|----------------------|
| DFlash  | Linear 4096 | 512  | 0.0006        | 4096            | 1.009 [0.981, 1.040] |
| DFlash  | MLP 4096    | 512  | 0.0006        | 4096            | 0.978 [0.961, 0.998] |
| DFlash  | Linear 4096 | 512  | 0.0002        | 4096            | 1.008 [0.985, 1.029] |
| DFlash  | Linear 4096 | 512  | 0.0018        | 4096            | 1.003 [0.978, 1.029] |
| DFlash  | MLP 4096    | 512  | 0.0002        | 2048            | 1.011 [0.986, 1.039] |
| DFlash  | MLP 4096    | 512  | 0.0018        | 4096            | 0.961 [0.939, 0.983] |
| EAGLE-3 | Dense       | 512  | 0.0006        | 1024            | 0.950 [0.924, 0.976] |
| EAGLE-3 | Linear 512  | 512  | 0.0006        | 2048            | 0.969 [0.948, 0.988] |
| EAGLE-3 | Linear 2048 | 512  | 0.0006        | 2048            | 0.964 [0.922, 1.003] |
| EAGLE-3 | MLP 2048    | 512  | 0.0006        | 2048            | 0.968 [0.942, 0.996] |
| EAGLE-3 | Dense       | 2048 | 0.0006        | 2048            | 0.980 [0.969, 0.992] |
| EAGLE-3 | MLP 2048    | 2048 | 0.0006        | 4096            | 1.019 [0.995, 1.046] |

Lower feature-validation loss does not consistently predict faster decoding. For EAGLE-3’s dense  $N = 512$  fit, the selected 1,024-update checkpoint retains 94.99% of endpoint throughput, with interval [92.39, 97.59]%. For the DFlash width-4096 MLP at  $N = 512$  and learning rate 0.0006, the selected 4,096-update checkpoint retains 97.77%, with interval [96.09, 99.76]%. Other intervals cross equal throughput. These results support treating feature regression as a surrogate and measuring decoding when selecting a connector. They do not establish full-answer quality or justify selection on untouched evaluation data.

Table 38: Observed throughput ranges across three fitting seeds in a separate 16-question, 256-token development campaign. Selected uses each fit’s minimum saved feature-validation checkpoint. These ranges describe the three measured fits and runtime variation, not confidence intervals.

| Mapper      | $N$  | Learning rate | Endpoint tok/s range | Selected tok/s range |
|-------------|------|---------------|----------------------|----------------------|
| Dense       | 512  | 0.0006        | 184.34–184.68        | 183.36–184.58        |
| Dense       | 2048 | 0.0006        | 186.11–189.22        | 186.11–189.22        |
| Linear 1024 | 512  | 0.0006        | 179.47–181.23        | 179.47–181.23        |
| Linear 4096 | 2048 | 0.0002        | 185.36–187.81        | 185.36–187.81        |
| MLP 4096    | 2048 | 0.0002        | 183.92–184.78        | 183.92–184.78        |
| MLP 4096    | 512  | 0.0002        | 174.53–177.30        | 175.57–177.56        |

All six settings in Table 38 use the same cached features and seeds 1729, 1730 and 1731. The dense  $N = 512$  endpoints span 184.34–184.68 tokens/s. The width-4096  $N = 2048$  MLP improves on its  $N = 512$  counterpart, while the dense connector remains competitive. These short development runs do not establish the minimum sufficient data count or distinguish small architecture gaps from runtime variation.

### C.3 LONGER DEVELOPMENT OUTPUTS FOR SMALL-DATA MAPPERS

We evaluate nine saved DFlash connectors on 128 paired MATH development questions with a 2,048-token cap, greedy decoding and the same source, target and native-drafter references. These exposed questions provide a larger development comparison than the 16-question, 256-token screen. They are separate from any untouched confirmation set. The continued width-512 fits reuse the same 2,048 distinct training examples.

Table 39: Capped development outcomes for small-data DFlash fitting. Throughput retention uses the dense  $N = 2048$ , 8,192-update connector. Intervals use 10,000 paired request bootstrap samples. Correct counts scored answers. Cap counts outputs reaching 2,048 tokens, which can include EOS exactly at the limit and are not exact truncation counts.

| Method         | $N$  | Updates | Tok/s  | Dense retained (%) [95% CI] | Correct | Cap |
|----------------|------|---------|--------|-----------------------------|---------|-----|
| AR             | –    | –       | 37.72  | 19.04 [18.23, 19.90]        | 104/128 | 11  |
| Native drafter | –    | –       | 204.57 | 103.27 [102.35, 104.17]     | 105/128 | 9   |
| Source reuse   | –    | –       | 120.96 | 61.06 [60.53, 61.59]        | 105/128 | 9   |
| Dense          | 512  | 8192    | 193.68 | 97.78 [97.09, 98.47]        | 105/128 | 9   |
| Dense          | 2048 | 8192    | 198.09 | 100.00 [100.00, 100.00]     | 105/128 | 9   |
| Linear 1024    | 512  | 8192    | 188.19 | 95.00 [94.14, 95.85]        | 105/128 | 9   |
| Linear 4096    | 2048 | 8192    | 196.25 | 99.07 [98.50, 99.60]        | 105/128 | 9   |
| MLP 4096       | 2048 | 8192    | 190.15 | 95.99 [95.30, 96.70]        | 105/128 | 9   |
| Linear 512     | 2048 | 8192    | 164.61 | 83.10 [81.81, 84.38]        | 105/128 | 9   |
| Linear 512     | 2048 | 32768   | 167.29 | 84.46 [83.15, 85.71]        | 105/128 | 9   |
| MLP 512        | 2048 | 8192    | 154.39 | 77.94 [76.50, 79.35]        | 105/128 | 9   |
| MLP 512        | 2048 | 32768   | 157.68 | 79.60 [78.07, 81.06]        | 105/128 | 9   |

Dense fitting on 512 examples retains 97.78% of the dense  $N = 2048$  throughput, with interval [97.09, 98.47]%. This establishes a lower-data performance point in this setting, but does not establish that 512 is minimal. The width-1024 linear mapper at  $N = 512$  retains 95.00%, with interval [94.14, 95.85]%, leaving its 95% boundary unresolved. The width-4096 linear mapper is slightly below the dense connector in this larger sample, so the earlier short-screen point ranking should not be interpreted as an established architecture advantage.

All nine connectors score 105 of 128 answers, compared with 104 for AR. Their paired accuracy-difference intervals are  $[-0.03125, 0.046875]$ . These intervals do not establish a one-percentage-point noninferiority margin. Nine connector outputs and eleven AR outputs reach the token cap. The observed scores therefore concern the configured capped evaluation, not uncapped answer quality. Longer width-512 fitting gives modest throughput increases while remaining below the wider or dense connectors. This comparison does not treat additional updates as additional data.

#### C.4 EAGLE-3 LONGER-OUTPUT CAPACITY COMPARISON

Two disjoint 64-question shards evaluate eight saved EAGLE-3 maps and three shared controls on the same 128 exposed MATH development questions. All 1,408 expected method/request pairs are present. The 2,048-token cap, fitting checkpoints, source revisions and pinned scorer remain fixed across shards. An eight-request pilot precedes this evaluation.

Table 40: EAGLE-3 with an 8B target on 128 exposed development questions. Dense retention uses the  $N = 2048$  endpoint. Intervals are paired 95% request-bootstrap intervals conditional on the fits. Cap counts all outputs reaching 2,048 tokens. It is not an exact truncation count.

| Method         | $N$  | Tokens/s | Dense retained (%) [95% CI] | Correct | Cap |
|----------------|------|----------|-----------------------------|---------|-----|
| AR             | —    | 36.32    | 38.19 [37.14, 39.25]        | 104/128 | 11  |
| Native drafter | —    | 94.89    | 99.78 [98.86, 100.80]       | 105/128 | 7   |
| Source reuse   | —    | 66.80    | 70.24 [69.69, 70.84]        | 105/128 | 7   |
| Dense          | 512  | 93.47    | 98.28 [97.59, 99.01]        | 105/128 | 7   |
| Dense          | 2048 | 95.10    | 100.00 [100.00, 100.00]     | 105/128 | 7   |
| Linear 2048    | 512  | 93.85    | 98.69 [98.08, 99.31]        | 105/128 | 7   |
| Linear 2048    | 2048 | 93.59    | 98.41 [97.76, 99.12]        | 105/128 | 7   |
| MLP 2048       | 512  | 84.67    | 89.03 [88.24, 89.87]        | 105/128 | 7   |
| MLP 2048       | 2048 | 89.25    | 93.85 [93.27, 94.46]        | 105/128 | 7   |
| Linear 512     | 2048 | 84.91    | 89.29 [88.54, 90.09]        | 105/128 | 7   |
| MLP 512        | 2048 | 79.76    | 83.87 [82.94, 84.84]        | 105/128 | 7   |

Dense  $N = 512$  retains 98.28% of dense  $N = 2048$  throughput. Linear-2048 at  $N = 512$  retains 98.69%, while the two MLP-2048 fits retain 89.03% and 93.85% at  $N = 512$  and  $N = 2048$ . The width-512 endpoints are slower. Every speculative method scores 105/128 with seven cap-length outputs. AR scores 104/128 with eleven. All speculative methods have identical per-question correctness in this sample, but the conservative paired 95% difference interval is still  $[-3.37, 3.37]$  percentage points. Against AR, dense  $N = 2048$  has four beneficial and three adverse correctness discordances. Equal or close scores therefore do not establish tight noninferiority or uncapped answer quality. The observed small-data retention replicates across the two drafter families under these development conditions, not across independent benchmark samples.

#### C.5 INPUT-ONLY TASK COMPLEXITY

We stratify the same 128 development requests using the benchmark’s original difficulty levels and subjects, together with shared tokenized input length. Group definitions are fixed before computing subgroup outcomes. They do not use candidate success, generated output length or latency. This is a retrospective analysis of exposed development data, not a new confirmatory test. All twelve methods and all requests remain in the analysis.



Table 41: Throughput retained from dense  $N = 2048$  within each MATH difficulty group, in percent with descriptive paired 95% intervals. All fits use 8,192 updates except the continued width-512 fits, which use 32,768 updates on the same 2,048 examples. For fitted maps, unspecified  $N$  is 2,048. Intervals do not adjust for multiple subgroup comparisons or model selection.

| Setting                 | Levels 1–2 ( $n = 42$ ) | Level 3 ( $n = 21$ ) | Levels 4–5 ( $n = 65$ ) |
|-------------------------|-------------------------|----------------------|-------------------------|
| AR                      | 20.1 [18.6, 21.5]       | 19.0 [16.5, 22.1]    | 18.7 [17.8, 19.7]       |
| Native drafter          | 101.4 [99.1, 104.1]     | 104.2 [102.0, 106.6] | 103.7 [102.8, 104.6]    |
| Source reuse            | 60.6 [59.2, 61.8]       | 60.7 [59.2, 62.1]    | 61.3 [60.6, 61.9]       |
| Dense, $N = 512$        | 97.4 [96.1, 98.6]       | 97.9 [96.1, 100.2]   | 97.9 [97.0, 98.8]       |
| Dense, $N = 2048$       | 100.0 [100.0, 100.0]    | 100.0 [100.0, 100.0] | 100.0 [100.0, 100.0]    |
| Linear 1024, $N = 512$  | 94.5 [93.2, 95.9]       | 94.8 [93.4, 96.5]    | 95.2 [94.0, 96.4]       |
| Linear 4096, $N = 2048$ | 98.9 [97.9, 99.8]       | 100.0 [98.6, 101.5]  | 98.9 [98.2, 99.6]       |
| MLP 4096, $N = 2048$    | 95.5 [94.3, 96.6]       | 96.8 [94.9, 98.7]    | 96.0 [95.1, 96.9]       |
| Linear 512              | 82.6 [79.3, 85.7]       | 84.5 [81.6, 86.8]    | 83.0 [81.5, 84.5]       |
| Linear 512, continued   | 84.1 [80.6, 87.5]       | 86.2 [84.0, 88.2]    | 84.2 [82.7, 85.7]       |
| MLP 512                 | 76.5 [73.4, 79.5]       | 79.9 [76.8, 82.5]    | 78.0 [76.2, 79.7]       |
| MLP 512, continued      | 78.4 [74.6, 82.0]       | 79.8 [77.0, 82.0]    | 80.0 [78.1, 81.7]       |

Dense  $N = 512$  retains 97.4–97.9% of dense  $N = 2048$  throughput across the three difficulty groups. The width-1024 linear map stays near the 95% boundary, while MLP-4096 retains 95.4–96.8%. The smaller width-512 maps remain slower in every group. These results describe the tested capacities and workloads. They do not establish an optimal size across settings.

For levels 1–2, level 3 and levels 4–5, respectively, AR answers 38/42, 19/21 and 47/65 questions correctly. Every mapper answers 38/42, 18/21 and 49/65 correctly. All eleven AR cap hits occur in the highest difficulty group, as do eight of the nine mapper cap hits. Difficulty therefore coincides with changes in capped answer quality and output length. It is not a controlled intervention on decoding cost.

Input-length bins of at most 64, 65–128, 129–256 and more than 256 tokens contain 52, 57, 15 and 4 requests. In the final four-request group, dense  $N = 512$  retains 94.5% [91.3, 96.4]% and linear-1024 retains 90.8% [88.2, 94.4]% of dense  $N = 2048$  throughput. The small sample and multiple comparisons make this a signal for replication, not evidence of a causal length effect or a new tuning rule. All seven original subject strata, input assignments, method outcomes and conservative paired accuracy intervals are retained in the accompanying artifacts. No additional generation or fitting is used for this analysis.

## C.6 CAPACITY REPLICATION WITH A 14B TARGET

We repeat the fixed-data capacity comparison with Qwen3-14B for both DFlash and EAGLE-3. Each family has fourteen primary fits: dense, factored linear and GELU MLP widths 512, 1,024 and 4,096 at each of  $N = 512$  and  $N = 2048$ . Two additional dense fits use seed 1730. All receive 8,192 batch-four updates at learning rate  $6 \times 10^{-4}$ , with no L2 penalty or weight decay. Validation uses the same 1,024 records within each family. Training diagnostics use a 256-record prefix. The five target taps give 25,600 input dimensions and the output has 2,560 dimensions. Dense therefore has 65.54M parameters, versus 14.42M, 28.84M and 115.34M at the three matched factorized/MLP widths. Width 4,096 does not increase the maximum rank of a linear function beyond the output dimension.

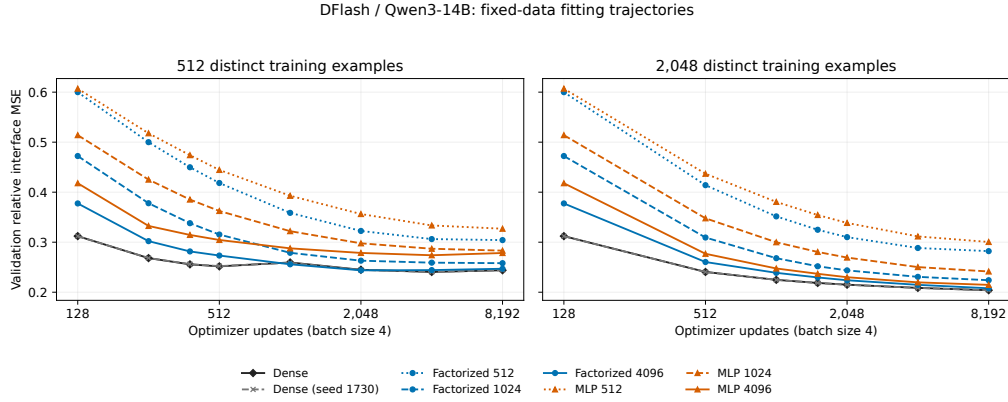


Figure 12: DFlash with Qwen3-14B. Validation error across the saved training trajectory, with all capacities and both dense seeds retained. The decoding endpoint remains fixed at 8,192 updates.

Table 42: DFlash-14B fixed endpoints. Throughput retention uses dense  $N = 2048$ , seed 1729. Intervals are paired 95% request-bootstrap intervals on sixteen exposed questions capped at 256 output tokens. They exclude fitting-seed and selection uncertainty.

| Mapper            | $N$  | Params (M) | Train / val.  | Tokens/s | Retained (%)         |
|-------------------|------|------------|---------------|----------|----------------------|
| Dense             | 512  | 65.54      | 0.153 / 0.244 | 123.38   | 96.0 [92.3, 99.3]    |
| Dense (seed 1730) | 512  | 65.54      | 0.153 / 0.244 | 122.08   | 94.9 [91.5, 98.1]    |
| Linear 512        | 512  | 14.42      | 0.251 / 0.304 | 99.16    | 77.1 [72.6, 81.8]    |
| Linear 1024       | 512  | 28.84      | 0.180 / 0.258 | 117.39   | 91.3 [87.7, 95.1]    |
| Linear 4096       | 512  | 115.34     | 0.155 / 0.247 | 121.65   | 94.6 [91.4, 97.9]    |
| MLP 512           | 512  | 14.42      | 0.277 / 0.327 | 92.11    | 71.6 [68.1, 75.8]    |
| MLP 1024          | 512  | 28.84      | 0.208 / 0.283 | 109.02   | 84.8 [79.7, 90.3]    |
| MLP 4096          | 512  | 115.34     | 0.140 / 0.278 | 112.85   | 87.8 [83.5, 92.0]    |
| Dense             | 2048 | 65.54      | 0.188 / 0.204 | 128.58   | 100.0 [100.0, 100.0] |
| Dense (seed 1730) | 2048 | 65.54      | 0.188 / 0.204 | 129.55   | 100.8 [99.5, 102.1]  |
| Linear 512        | 2048 | 14.42      | 0.274 / 0.282 | 105.53   | 82.1 [76.9, 87.4]    |
| Linear 1024       | 2048 | 28.84      | 0.212 / 0.224 | 127.67   | 99.3 [96.5, 102.3]   |
| Linear 4096       | 2048 | 115.34     | 0.193 / 0.208 | 126.96   | 98.7 [96.5, 101.2]   |
| MLP 512           | 2048 | 14.42      | 0.292 / 0.300 | 96.46    | 75.0 [70.9, 79.2]    |
| MLP 1024          | 2048 | 28.84      | 0.227 / 0.241 | 121.21   | 94.3 [90.9, 98.1]    |
| MLP 4096          | 2048 | 115.34     | 0.174 / 0.215 | 122.32   | 95.1 [92.6, 98.2]    |

With DFlash, the wide MLP at  $N = 2048$  has lower training error than dense, but higher validation error and lower measured throughput. The width-1024 linear map nearly matches dense throughput with fewer parameters. These comparisons use fixed endpoints, not checkpoints chosen after inspecting decoding. Every method reaches the output cap on these sixteen questions, so the short scores do not establish full-answer quality. The runtime includes AR and source-reuse controls. No native 14B DFlash checkpoint is substituted from another target.

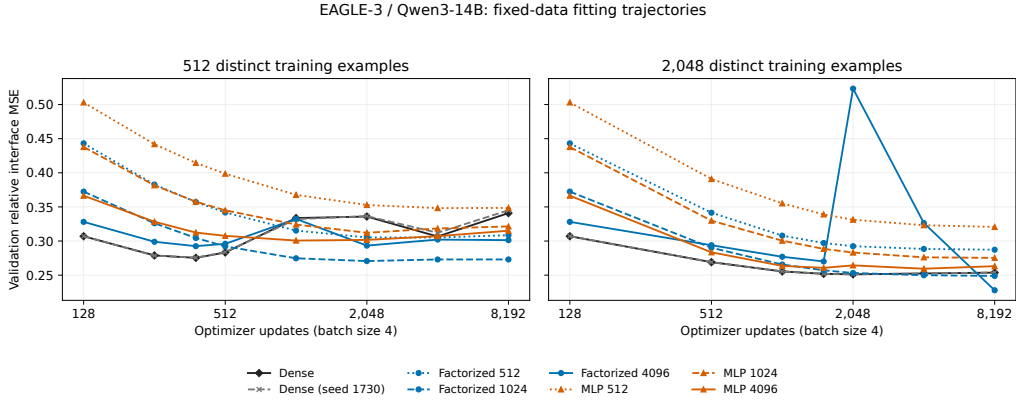


Figure 13: EAGLE-3 with Qwen3-14B under the same data-count and update budgets. The width-4096 linear trajectory retains its intermediate validation spike. Curves show optimization behavior, not a ranking of decoding performance or a selected stopping rule.

Table 43: EAGLE-3-14B fixed endpoints under its own runtime. Retention and intervals use the same within-family protocol as Table 42. Absolute throughput across the two tables is not an algorithm-only comparison.

| Mapper            | $N$  | Params (M) | Train / val.  | Tokens/s | Retained (%)         |
|-------------------|------|------------|---------------|----------|----------------------|
| Dense             | 512  | 65.54      | 0.234 / 0.341 | 62.24    | 94.4 [91.7, 97.2]    |
| Dense (seed 1730) | 512  | 65.54      | 0.238 / 0.345 | 63.32    | 96.1 [92.7, 99.5]    |
| Linear 512        | 512  | 14.42      | 0.256 / 0.308 | 56.92    | 86.4 [83.5, 89.4]    |
| Linear 1024       | 512  | 28.84      | 0.197 / 0.273 | 62.50    | 94.8 [91.9, 98.0]    |
| Linear 4096       | 512  | 115.34     | 0.226 / 0.301 | 60.90    | 92.4 [89.8, 95.1]    |
| MLP 512           | 512  | 14.42      | 0.302 / 0.349 | 53.15    | 80.6 [77.4, 84.1]    |
| MLP 1024          | 512  | 28.84      | 0.253 / 0.321 | 59.45    | 90.2 [87.3, 92.8]    |
| MLP 4096          | 512  | 115.34     | 0.224 / 0.315 | 52.64    | 79.9 [76.6, 83.5]    |
| Dense             | 2048 | 65.54      | 0.231 / 0.254 | 65.91    | 100.0 [100.0, 100.0] |
| Dense (seed 1730) | 2048 | 65.54      | 0.231 / 0.254 | 66.45    | 100.8 [99.3, 102.4]  |
| Linear 512        | 2048 | 14.42      | 0.279 / 0.287 | 58.92    | 89.4 [86.0, 92.7]    |
| Linear 1024       | 2048 | 28.84      | 0.236 / 0.249 | 63.41    | 96.2 [94.6, 97.8]    |
| Linear 4096       | 2048 | 115.34     | 0.216 / 0.228 | 65.61    | 99.5 [97.5, 101.9]   |
| MLP 512           | 2048 | 14.42      | 0.311 / 0.321 | 56.54    | 85.8 [82.2, 89.3]    |
| MLP 1024          | 2048 | 28.84      | 0.259 / 0.275 | 61.76    | 93.7 [90.5, 97.0]    |
| MLP 4096          | 2048 | 115.34     | 0.227 / 0.263 | 64.02    | 97.1 [94.9, 99.4]    |

EAGLE-3 differs from DFlash in its fitting dynamics. At  $N = 512$ , both dense seeds have their lowest saved validation error at 384 updates, followed by substantial increases. At  $N = 2048$ , the wide linear endpoint has lower validation error than dense, but nearly equal decoding throughput. Its intermediate spike also shows that the shared learning rate does not produce uniformly smooth training. We complete all four previously declared lower-rate checks at 8,192 updates, preserving the original records, feature cache and seed. Lower rates improve every endpoint’s feature-validation error and avoid the wide-linear spike at the saved checkpoints (Table 44). Paired decoding compares each new map with its original-rate counterpart. The best-fitting dense map is not the fastest: rate 0.00006 gives validation error 0.226, versus 0.250 at 0.0002, but the latter has higher throughput. Thus optimization affects the capacity comparison, while feature error remains an imperfect decoding surrogate.

| Mapper      | Records | LR     | Original val. | New val. | Ratio (95% CI)      |
|-------------|---------|--------|---------------|----------|---------------------|
| Dense       | 512     | 0.0002 | 0.341         | 0.250    | 1.069 [1.042,1.098] |
| Dense       | 512     | 6e-05  | 0.341         | 0.226    | 1.037 [1.011,1.064] |
| Linear-4096 | 2,048   | 0.0002 | 0.228         | 0.213    | 1.017 [0.993,1.040] |
| MLP-4096    | 2,048   | 0.0002 | 0.263         | 0.214    | 1.044 [1.014,1.070] |

Table 44: Targeted EAGLE-3/14B learning-rate checks, one fitting seed. Original learning rate is 0.0006. Ratios compare the lower-rate map with the original map at matched architecture, records and 8,192 updates on eight exposed MATH questions, with paired 95% request intervals. All four checks are retained. Four of eight outputs reach the 512-token cap for every method, so these are development throughput results, not full-answer quality confirmation.

Full-answer quality, broader workloads and additional fitting seeds remain separate from this capped comparison.

### C.7 MATCHED WARM-TRAINING BUDGETS ON 512 EXAMPLES

We compare feature regression, connector cross-entropy (CE) and rank-32 drafter LoRA on the same 512-example pool. CE updates the connector while LoRA holds the initial connector fixed. Both use target-greedy teacher labels on the last 15 proposal positions. An equal three-rate development screen selects learning rate  $2 \times 10^{-5}$  for each adaptation family. Feature regression uses its declared rate  $6 \times 10^{-4}$ . The two LoRA seeds vary the update initialization and share the same initial mapper. They do not measure variation across initial mapper fits.

We measure 91.413 seconds of optimizer-update time for the 8,192-update feature baseline and declare budgets of 0.25, 1 and 4 times that cost. CE and LoRA begin with a 128-update mapper, whose measured 1.546 seconds of training counts toward each budget. Each worker stops after the first completed update reaching its remaining budget. Recorded overshoot is at most one update, ranging from 0.002 to 0.201 seconds across the twelve fits. The three four-GPU fitting jobs finish in 2m33s, 2m35s and 7m12s.

Table 45: Measured training costs and subsequent decoding in one common development campaign. Worker time includes model/cache setup, validation and export where performed, plus the measured initial mapper worker cost for CE and LoRA. It excludes Python startup and shared cache construction. Ratios compare against feature regression at the same warm budget.

| Budget | Objective     | Updates | Warm s | Worker s | Tok/s  | Feature ratio [95% CI] |
|--------|---------------|---------|--------|----------|--------|------------------------|
| 0.25×  | Feature MSE   | 2026    | 22.86  | 52.04    | 145.05 | 1.000 [1.000, 1.000]   |
| 0.25×  | Connector CE  | 106     | 22.94  | 91.89    | 111.04 | 0.766 [0.737, 0.791]   |
| 0.25×  | LoRA32 (1729) | 95      | 23.00  | 94.66    | 119.30 | 0.823 [0.786, 0.859]   |
| 0.25×  | LoRA32 (1730) | 94      | 22.86  | 95.20    | 119.52 | 0.824 [0.788, 0.860]   |
| 1×     | Feature MSE   | 8144    | 91.42  | 119.71   | 144.80 | 1.000 [1.000, 1.000]   |
| 1×     | Connector CE  | 474     | 91.58  | 161.04   | 109.60 | 0.757 [0.726, 0.786]   |
| 1×     | LoRA32 (1729) | 402     | 91.45  | 165.73   | 115.02 | 0.794 [0.757, 0.831]   |
| 1×     | LoRA32 (1730) | 407     | 91.50  | 164.61   | 115.49 | 0.798 [0.765, 0.830]   |
| 4×     | Feature MSE   | 32446   | 365.65 | 390.11   | 144.55 | 1.000 [1.000, 1.000]   |
| 4×     | Connector CE  | 1899    | 365.72 | 435.56   | 105.79 | 0.732 [0.699, 0.763]   |
| 4×     | LoRA32 (1729) | 1651    | 365.85 | 438.51   | 118.49 | 0.820 [0.786, 0.853]   |
| 4×     | LoRA32 (1730) | 1646    | 365.76 | 437.99   | 119.55 | 0.827 [0.793, 0.860]   |

Table 46: References measured within the same budget-decoding campaign. The initial mapper has 128 feature-regression updates. The 8,192-update reference is an existing saved checkpoint.

| Reference           | Tok/s  | Initial mapper ratio [95% CI] |
|---------------------|--------|-------------------------------|
| AR                  | 28.61  | 0.240 [0.207, 0.277]          |
| Native drafter      | 151.71 | 1.275 [1.205, 1.348]          |
| Source reuse        | 84.71  | 0.712 [0.684, 0.737]          |
| Initial mapper      | 119.01 | 1.000 [1.000, 1.000]          |
| Dense, 8192 updates | 145.82 | 1.225 [1.154, 1.302]          |

Feature regression reaches 145.05, 144.80 and 144.55 tokens/s at the three budgets. Relative to the quarter-budget fit, the 1-times and 4-times fits retain 99.83% [97.18, 102.53]% and 99.66% [97.35, 102.11]%, respectively. These intervals do not establish a benefit from longer fitting. Connector CE retains 73.2–76.6% of same-budget feature throughput, and the two LoRA seeds retain 79.4–82.7%. This result concerns the declared teacher-forced objective, initialization, rank and development-selected learning rates. It does not establish that other adaptation objectives or full drafter training fail.

Warm training is a marginal cost with cached features available. The historical full-cache extraction took 712.58 seconds and produced 292.8 GB. We reuse its first 512 training records and report the original construction cost separately. Dividing that cost by the record count would not measure a fresh 512-example extraction. The timed fits do not add training examples or repeat large-data scaling.

All seventeen methods decode the same sixteen exposed development questions with a 256-token cap. Tables 45 and 46 report descriptive paired request bootstrap intervals. They do not provide full-answer quality, untouched confirmation or corrections for selecting among multiple comparisons. Absolute throughput is interpreted within this common campaign and its deterministic training/runtime configuration.

#### C.8 DRAFT LENGTH AND INTERFACE CHOICES

The block-size experiment directly compares AR, native DFlash and the relay at lengths 8, 16 and 32 on 64 fixed requests. Block 16 gives the highest relay throughput, 210.15 tokens/s (Table 13). Block 8 retains more source progress proportionally but advances fewer tokens per cycle. Block 32 adds verification width while reducing accepted progress for the released block-16 drafter. This supports keeping its trained block length instead of maximizing the relative gain over source reuse.

The EAGLE-3 normalization comparison holds parameter count, data and 128-update budget fixed. Raw target features reduce relative context error from 0.413 to 0.266 and increase the source-relative throughput ratio from 1.035 to 1.237. The raw interface consumes feature magnitude, which input normalization removes. This gives a concrete reason for preserving scale in the selected map.

Appendix C reports the DFlash objective, ridge and normalization comparisons with their distinct fitting recipes.

## D EXTERNAL BASELINES IN THEIR PUBLIC RUNTIMES

Table 47: Mechanism and evaluation scope of the closest alternatives. All rows inherit pretrained models. “No new fit” for PARD refers to retargeting its released checkpoint, not its original training cost. DFlash and EAGLE-3 are the inherited drafter families rather than independent competitors to RelaySpec’s interface replacement.

| Method                                | New-target adaptation                        | Use of target features            | Evidence here                                      |
|---------------------------------------|----------------------------------------------|-----------------------------------|----------------------------------------------------|
| RelaySpec                             | Fit input map, freeze drafter                | Replace source conditioning       | Two drafter families, 8B/14B targets               |
| PARD (An et al., 2026)                | No new fit for released drafter              | Target-independent proposals      | Public runtime, 128-question capped quality        |
| SD <sup>2</sup> (Berdoz et al., 2026) | Fit steering in tested frozen-drafter option | Inject steering within drafter    | Rate and epoch sweeps, 128-question capped quality |
| Connector CE                          | Fit input map with token CE                  | Same replacement interface        | Common-runtime matched warm budgets                |
| Drafter LoRA                          | Fit initial map, then LoRA with map frozen   | Adapt feature-conditioned drafter | Common-runtime matched warm budgets, two seeds     |

We evaluate SD<sup>2</sup>’s public frozen-drafter option and the released PARD parallel drafter on sixteen exposed MATH development requests with a 256-token cap. Each system has a correct cached AR control using its own target runtime and identical pretokenized inputs. These are short development comparisons. Below we separately evaluate capped answer quality. No untouched confirmation is included in these measurements.

SD<sup>2</sup> trains 402,665,472 steering parameters with merged guidance from target layers 3, 16 and 33 into all drafter layers. Inherited target and drafter weights remain frozen. Each of six objective/rate settings sees 512 distinct records once, using 128 batch-four updates and 95,835 non-padding supervised positions. Records are capped at 192 tokens. Training takes 47.0–48.6 seconds per fit. The initial TVD and KL settings also have exact fitting replicas, whose costs remain in the experiment record. This supervision differs from the feature-regression and last-15-position adaptation objectives in the preceding study.

Table 48: One-epoch SD<sup>2</sup> development comparison. Intervals resample paired requests 10,000 times and do not adjust for rate selection. Exact AR counts identical capped token sequences.

| Setting                 | Tok/s | AR ratio [95% CI]    | Independent ratio [95% CI] | Exact AR |
|-------------------------|-------|----------------------|----------------------------|----------|
| Matched AR              | 12.64 | 1.000 [1.000, 1.000] | 0.671 [0.622, 0.719]       | 16/16    |
| Independent drafter     | 18.83 | 1.490 [1.391, 1.607] | 1.000 [1.000, 1.000]       | 6/16     |
| Zero guidance           | 18.56 | 1.469 [1.371, 1.584] | 0.985 [0.983, 0.988]       | 6/16     |
| TVD, $10^{-5}$          | 18.38 | 1.454 [1.360, 1.567] | 0.976 [0.955, 0.997]       | 4/16     |
| KL, $10^{-5}$           | 18.17 | 1.438 [1.341, 1.552] | 0.965 [0.937, 0.993]       | 9/16     |
| TVD, $4 \times 10^{-6}$ | 18.52 | 1.466 [1.370, 1.574] | 0.984 [0.961, 1.006]       | 6/16     |
| TVD, $2 \times 10^{-5}$ | 18.47 | 1.462 [1.365, 1.570] | 0.981 [0.953, 1.011]       | 6/16     |
| KL, $4 \times 10^{-6}$  | 18.66 | 1.477 [1.384, 1.588] | 0.991 [0.962, 1.016]       | 5/16     |
| KL, $2 \times 10^{-5}$  | 18.42 | 1.458 [1.363, 1.569] | 0.978 [0.955, 1.003]       | 10/16    |

The public SD<sup>2</sup> evaluation configuration uses FP16 target storage, BF16 drafter and steering storage, and BF16 autocast under Transformers 4.52.4. We verify original FP32 training weights before conversion and separately fingerprint the BF16 inference tensors. The best fitted point, KL at  $4 \times 10^{-6}$ , retains 99.11% [96.19, 101.62]% of independent-drafter throughput. This screen does not establish a steering gain. We freeze this rate before checking additional epochs on the same small pool.

Table 49: Released PARD with zero new-target fitting, Transformers 4.51.3, BF16 weights, eager attention and static caches. The released checkpoint carries inherited proposer training.

| Setting          | Tok/s | AR ratio [95% CI]    | Exact AR |
|------------------|-------|----------------------|----------|
| Matched eager AR | 31.27 | 1.000 [1.000, 1.000] | 16/16    |
| Released PARD    | 98.60 | 3.153 [2.831, 3.558] | 4/16     |

Every  $SD^2$  committed token matches its actual verifier decision, and immediate versus deferred observation preserves tokens and acceptance. Each timed PARD request has a separate verifier-observed replica with identical raw output and acceptance counts. PARD’s measured generation omits that added observer. Both timers include prefills and cache setup, preserve raw EOS/cap overshoot and exclude detokenization. Earlier failed exact-AR gates remain failed. Controlled cache, query and attention replays expose changes in rounded target scores, while future-query perturbations preserve earlier logits in the tested calls. The sequence differences in these tables require separate task-quality evidence. Absolute speeds across these public runtimes do not isolate algorithm effects, and shared model residency does not measure isolated serving memory.

#### D.1 ADDITIONAL EPOCHS AND CAPPED ANSWER QUALITY

After the six-rate screen, we fix KL and the selected learning rate and fit for 1, 2, 4 and 8 epochs on the same 512 examples. These are four fresh fits, each with its own learning-rate decay horizon, rather than checkpoints from one training trajectory. The one-epoch fit exactly reproduces the selected earlier training fingerprint. All four endpoints are evaluated together on the same sixteen development requests, with the public  $SD^2$  runtime and matched controls.

Table 50:  $SD^2$  epoch budget at fixed 512 distinct examples. Training KL is the mean over updates in each final epoch. It is not held-out loss. Independent-drafter throughput is 18.75 tokens/s. Intervals resample paired requests and do not adjust for model selection.

| Epochs | Fit seconds | Final-epoch training KL | Tok/s | Independent ratio [95% CI] |
|--------|-------------|-------------------------|-------|----------------------------|
| 1      | 48.44       | 0.3923                  | 18.56 | 0.9900 [0.9607, 1.0149]    |
| 2      | 97.01       | 0.2787                  | 18.45 | 0.9841 [0.9639, 1.0033]    |
| 4      | 193.81      | 0.2027                  | 18.39 | 0.9808 [0.9555, 1.0039]    |
| 8      | 388.08      | 0.1254                  | 18.26 | 0.9738 [0.9488, 0.9997]    |

Training KL decreases from 0.3923 to 0.1254, while every fitted endpoint has a throughput point below the independent drafter. Additional epochs do not establish a decoding improvement in this fixed-pool setting. The highest-throughput fitted endpoint remains one epoch. We freeze this checkpoint before longer quality evaluation. This result concerns the tested frozen-drafter configuration, supervision and pool, and does not establish a general failure of  $SD^2$  or its other training options.

Table 51: PARD capped answer quality on 128 exposed MATH development requests, after an eight-request pilot passed. Both methods use the same 2048-token cap and public BF16 eager runtime. Cap hits include all outputs reaching the cap. None ends in EOS exactly at the cap.

| Setting          | Tok/s  | AR ratio [95% CI]    | Correct | Cap hits |
|------------------|--------|----------------------|---------|----------|
| Matched eager AR | 31.67  | 1.000 [1.000, 1.000] | 103/128 | 11       |
| Released PARD    | 105.63 | 3.335 [3.222, 3.451] | 105/128 | 8        |

PARD has five beneficial and three adverse correctness discordances against AR, a difference of +1.56 percentage points. A conservative paired 95% interval is  $[-6.34, 9.30]$  points: form exact 97.5% Clopper-Pearson intervals for each discordance probability and combine their opposite endpoints using a Bonferroni bound. This construction requires IID request pairs and a fixed candidate. It does not establish a one-percentage-point noninferiority margin. The descriptive paired bootstrap interval

is  $[-2.34, 5.47]$  points. Sparse or absent discordances can make such intervals misleadingly narrow. These external-baseline measurements use development questions.

PARD matches AR’s full capped token sequence on 28 of 128 requests. Its mean output length is 664.5 tokens versus AR’s 690.4, so the 3.34-fold token-throughput ratio differs from the 3.47-fold request-time ratio. Every measured PARD request retains a separate, exactly matching verifier-observed replica. These are development outcomes under the specified runtime and scorer, not untouched confirmation or an isolated algorithm ranking across runtimes.

Table 52: Selected one-epoch  $SD^2$  on the same 128 exposed MATH development requests, with a 2048-token cap. The public mixed-precision runtime and matched AR control are retained. Both have 11 cap-length outputs, none with EOS exactly at the cap.

| Setting         | Tok/s | AR ratio [95% CI]    | Correct | Cap hits |
|-----------------|-------|----------------------|---------|----------|
| Matched AR      | 12.48 | 1.000 [1.000, 1.000] | 103/128 | 11       |
| Selected $SD^2$ | 19.92 | 1.596 [1.560, 1.632] | 103/128 | 11       |

$SD^2$  and matched AR both score 103 of 128, with three beneficial and three adverse correctness discordances. The conservative paired 95% interval is  $[-7.04, 7.04]$  percentage points, which does not establish the one-point noninferiority margin. The descriptive paired bootstrap interval is  $[-3.91, 3.91]$  points. Full capped sequences agree on 42 requests. Mean output lengths are 679.1 versus 680.9 tokens, so the 1.60-fold token-throughput and request-time ratios are close. Both shards retain the same selected training and inference fingerprints, and saved answers reproduce under the pinned scorer.

The public physical-buffer limit can end an untimed warmup before its token cap. We record this termination only during warmup. Measured requests must still reach EOS or the cap and pass all actual-verifier checks. A replacement eight-request pilot passed before the complete quality run. Failed preliminary attempts remain excluded from the reported timings. These results concern the tested frozen-drafter configuration and its public runtime, not all  $SD^2$  training options.

## E ACCEPTANCE AND NUMERICAL AGREEMENT

A retention value of 90% means the reused drafter produces 90% as many tokens per second as the target’s own drafter. It is not the fraction of speedup above AR. We also report the ratio of summed AR request time to summed relay request time. This differs from a throughput ratio when output lengths differ.

Acceptance explains how speed is obtained. We report the pooled number of tokens advanced per proposal-verification cycle, including the implementation’s target-supplied progress. This is distinct from the fraction of proposed draft tokens accepted. Appendix E gives the position-wise curves and precise counting convention.

Let  $c_R$  be the average relay cycle time and  $a_R$  its average progress. Its decode time per output token is approximately  $c_R/a_R$ . If  $c_A$  is AR time per token, then the decode-only approximation is

$$\text{AR-relative decode speed} \approx \frac{a_R c_A}{c_R}. \quad (4)$$

Removing source conditioning reduces cycle cost. Imperfect feature matching may reduce progress. The map is useful when the saved computation outweighs the extra cycles. We measure end-to-end speed directly and use this approximation to interpret the component measurements.

We pool all recorded proposal cycles rather than averaging each request’s mean. This weights cycles equally. DFlash’s recorded length is one already target-selected position plus the accepted draft prefix. EAGLE-3’s length follows its shared decoder’s progress count. These quantities measure work advanced per cycle, including target-supplied progress. A proposed-token acceptance fraction requires separating those positions and accounting for the number proposed.



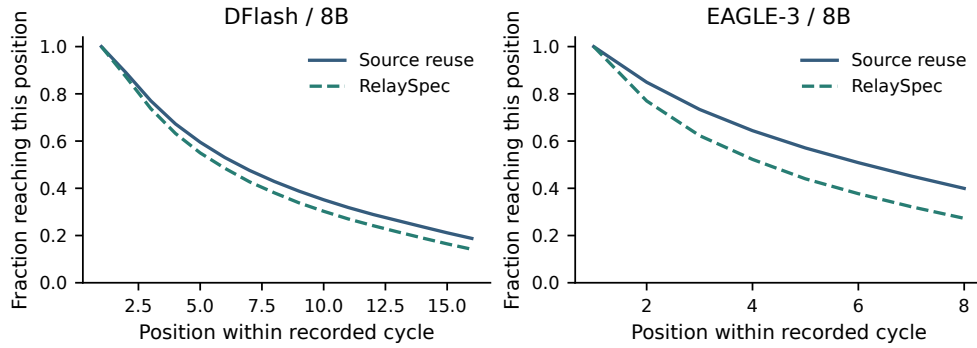


Figure 14: Historical 8B source-reuse comparisons. Each curve gives the fraction of recorded cycles reaching each position. Summing the curve values gives mean recorded progress per cycle. Solid and dashed lines show source reuse and RelaySpec, respectively.

The main MATH-500 table reports measured full-sequence agreement with AR. A fresh diagnostic selects eight prompts with early differences and saves target inputs, cache lengths, positions and the two leading logits at every call. The DFlash paths share the same first differing choice across native, source and relay drafting. The observed BF16 logits include ties or changed ordering with gaps up to 0.25 at those positions. Such traces identify numerical score differences in the selected cases. They do not establish that all mismatches have the same cause. The task-quality intervals in Table 2 remain the empirical quality evidence.

## F MATCHED-COUNT TRAINING COMPOSITION

We compare two 2,048-record DFlash-8B fitting pools. The math-only arm uses NuminaMath records. The mixed arm replaces half with 1,024 Dolly general-instruction records.<sup>2</sup> Exact and lexical filters exclude matches to fitting validation, recorded evaluation prompts and the confirmation and reserve sets. The common validation set contains 1,024 math and 1,024 general-instruction records. Its cached feature files are identical across arms.

Each arm fits dense, linear-512 and MLP-512 maps at seed 1729, plus a second dense seed, 1730. Each fit uses 8,192 batch-four updates, learning rate 0.0006, no regularization and the same relative interface objective. We retain the 2,048-update checkpoint as a fitting diagnostic, but preselect the 8,192-update endpoint for decoding rather than selecting by validation. Four independent fits run simultaneously on four L40S GPUs. Feature extraction took 111.4 seconds for the math cache and 107.1 for the mixed cache. The full fitting allocations took 4m36s and 4m21s, respectively. Extraction and fitting are separate costs.

Equal record counts and updates do not imply equal token exposure. With a 192-token record cap, the math pool contains 385,492 non-padding tokens and the mixed pool 316,518. Sixteen passes expose 6,167,872 and 5,064,288 tokens, respectively. The experiment therefore measures composition at matched records and updates, not matched tokens.

<sup>2</sup>[databricks/databricks-dolly-15k](https://databricks.com/databricks-dolly-15k), revision bdd27f4d94b9, CC-BY-SA 3.0. General instructions can include mathematical questions.

Table 53: Composition fitting at the preselected 8,192-update endpoint. Validation columns use the same held-out features in both arms. Reported seconds cover per-worker data access, updates and training-log writes, excluding validation, export and setup. Training losses concern different input distributions and are not directly comparable across arms.

| Mapper         | Training | Train loss | Math val. | General val. | Fit seconds |
|----------------|----------|------------|-----------|--------------|-------------|
| Dense          | math     | 0.1695     | 0.1805    | 0.3329       | 177.0       |
| Linear-512     | math     | 0.2629     | 0.2685    | 0.4719       | 114.5       |
| MLP-512        | math     | 0.2812     | 0.2878    | 0.4928       | 110.3       |
| Dense (seed 2) | math     | 0.1695     | 0.1805    | 0.3328       | 173.6       |
| Dense          | mixed    | 0.1854     | 0.1990    | 0.2234       | 167.0       |
| Linear-512     | mixed    | 0.3027     | 0.3015    | 0.3293       | 97.1        |
| MLP-512        | mixed    | 0.3221     | 0.3233    | 0.3479       | 110.2       |
| Dense (seed 2) | mixed    | 0.1852     | 0.1990    | 0.2231       | 152.7       |

Mixed training lowers dense general-instruction validation error by 32.9%, from 0.3329 to 0.2234, while raising math error by 10.2%, from 0.1805 to 0.1990. The second dense seed reproduces this tradeoff. At width 512, the linear and MLP maps show general-error reductions of 30.2% and 29.4%, with math-error increases of 12.3% in both cases. The matched linear map retains lower validation error than the MLP in both domains and both training arms. All three architectures improve in both validation domains between 2,048 and 8,192 updates.

**Fixed cross-domain decoding protocol.** Before observing endpoint results, we declare 32 development inputs from each of MATH-500, GSM8K, HumanEval and MT-Bench. The 128 initial inputs expand to 160 timed requests because MT-Bench includes two turns. All eight fixed endpoints run alongside AR, native DFlash and source reuse, producing 1,760 measured outputs capped at 2,048 tokens. A separate eight-input operational pilot completes in 2m58s before this full run. Methods rotate within requests in the same runtime. Ratios compare mixed and math-only fits of the same architecture and seed. Paired bootstrap intervals resample whole conversations for MT-Bench and individual requests elsewhere. They describe this development sample, without multiplicity adjustment or fitting-seed uncertainty.

The feature-domain tradeoff carries through to decoding, but depends on mapper capacity and task. Mixed dense training changes throughput by  $-1.2\%$  on MATH,  $+1.6\%$  on GSM8K,  $+2.9\%$  on HumanEval and  $+4.9\%$  on MT-Bench. The second dense seed gives  $-1.7\%$ ,  $+1.2\%$ ,  $+2.4\%$  and  $+4.7\%$ , respectively. The first seed’s MATH interval includes no change. Linear-512 and MLP-512 gain 11.6% and 10.8% on chat, but lose 6.2% and 7.8% on MATH and 2.6% and 4.1% on GSM8K. Dense remains faster than either smaller map on every task in both arms. General-domain feature improvement therefore supports useful workload-specific gains, not a uniform advantage from mixed training.

Every matched math/mixed pair produces identical generated token sequences on all 160 requests, including conversation turns. Thus these measured throughput gains do not arise from shorter outputs. All eight maps score 22/32 on MATH and 30/32 on GSM8K. AR scores 22/32 and 31/32. Four MATH outputs per method reach the cap, and none elsewhere does. The conservative paired accuracy-difference interval between matched arms is  $[-12.80, 12.80]$  percentage points on each math benchmark. Equal observed outcomes on this small sample do not establish tight population quality noninferiority. Code and conversation outputs remain unscored for task quality. The full decoding allocation took 52m30s.

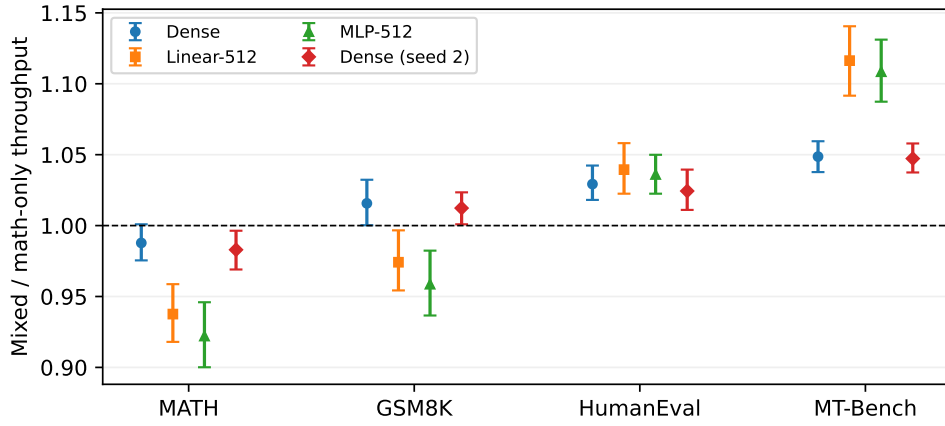


Figure 15: Effect of training composition on measured DFlash-8B throughput. Values above one favor mixed training. Marker shapes distinguish the four matched comparisons. Bars show paired 95% intervals. Code and conversation throughput does not establish output quality.

Table 54: All predeclared composition decoding comparisons. Throughput includes prompt processing. Ratios are mixed divided by math-only throughput, with paired 95% intervals. MT-Bench has 64 timed turns from 32 conversations. Each other task has 32 requests.

| Task      | Mapper         | Math tokens/s | Mixed tokens/s | Ratio (95% CI)      |
|-----------|----------------|---------------|----------------|---------------------|
| MATH      | Dense          | 201.1         | 198.6          | 0.988 [0.975,1.001] |
| GSM8K     | Dense          | 156.5         | 159.0          | 1.016 [1.000,1.032] |
| HumanEval | Dense          | 138.1         | 142.2          | 1.029 [1.018,1.042] |
| MT-Bench  | Dense          | 71.1          | 74.6           | 1.049 [1.038,1.059] |
| MATH      | Linear-512     | 165.4         | 155.1          | 0.938 [0.918,0.959] |
| GSM8K     | Linear-512     | 120.3         | 117.2          | 0.974 [0.954,0.997] |
| HumanEval | Linear-512     | 69.9          | 72.7           | 1.039 [1.022,1.058] |
| MT-Bench  | Linear-512     | 49.1          | 54.8           | 1.116 [1.092,1.141] |
| MATH      | MLP-512        | 155.6         | 143.5          | 0.922 [0.900,0.946] |
| GSM8K     | MLP-512        | 112.7         | 108.0          | 0.959 [0.937,0.982] |
| HumanEval | MLP-512        | 64.4          | 66.7           | 1.036 [1.022,1.050] |
| MT-Bench  | MLP-512        | 46.5          | 51.5           | 1.108 [1.087,1.131] |
| MATH      | Dense (seed 2) | 201.6         | 198.2          | 0.983 [0.969,0.996] |
| GSM8K     | Dense (seed 2) | 156.7         | 158.6          | 1.012 [1.001,1.023] |
| HumanEval | Dense (seed 2) | 138.0         | 141.4          | 1.024 [1.011,1.039] |
| MT-Bench  | Dense (seed 2) | 71.1          | 74.5           | 1.047 [1.037,1.058] |

Table 55: Composition evaluation: correct-answer and output-cap counts. MATH and GSM8K columns show correct answers followed by capped outputs, each out of 32. Code cap counts are out of 32 and chat cap counts out of 64 turns. HumanEval execution and MT-Bench answer quality are not scored in this experiment. Capped outputs remain in every aggregate.

| Method                | MATH correct / cap | GSM8K correct / cap | Code cap | Chat cap |
|-----------------------|--------------------|---------------------|----------|----------|
| AR                    | 22 / 4             | 31 / 0              | 0        | 0        |
| Native DFlash         | 22 / 4             | 30 / 0              | 0        | 0        |
| Source reuse          | 22 / 4             | 30 / 0              | 0        | 0        |
| Dense, math           | 22 / 4             | 30 / 0              | 0        | 0        |
| Dense, mixed          | 22 / 4             | 30 / 0              | 0        | 0        |
| Linear-512, math      | 22 / 4             | 30 / 0              | 0        | 0        |
| Linear-512, mixed     | 22 / 4             | 30 / 0              | 0        | 0        |
| MLP-512, math         | 22 / 4             | 30 / 0              | 0        | 0        |
| MLP-512, mixed        | 22 / 4             | 30 / 0              | 0        | 0        |
| Dense (seed 2), math  | 22 / 4             | 30 / 0              | 0        | 0        |
| Dense (seed 2), mixed | 22 / 4             | 30 / 0              | 0        | 0        |

### F.1 COMPOSITION WITH COMPACT INTERFACES

We separately screen two-layer interfaces using 512 math records or 256 math plus 256 general-instruction records, at 8,192 updates and seed 1729. Both use target layers [25,33]. Retargeted maps fit the source-drafter interface. Native students fit the released native projection and normalization. Thus this comparison changes calibration composition at fixed interface capacity and record count, but not necessarily equal token exposure.

The largest observed difference is on retargeted dialogue: mixed calibration retains 110.8% [107.8,112.5]% of math-only throughput. All seven other intervals include parity. Every paired output matches within these capped screens. This is a workload-specific exploratory finding from eight conversations and one fitting seed, not evidence that mixed calibration improves decoding across workloads or answer quality.

Table 56: Compact-interface calibration composition. Ratios compare both checkpoints within the same worker. Paired 95% bootstrap intervals use requests, or whole conversations for dialogue, and exclude fitting-seed and selection uncertainty. Code and math screens use 128-token caps. dialogue uses 256 tokens per turn.

| Interface  | Workload | Units | Mixed / math throughput (95% CI) |
|------------|----------|-------|----------------------------------|
| Native     | Code     | 32    | 100.6% [98.6, 102.7]             |
| Native     | Dialogue | 8     | 101.1% [98.8, 102.7]             |
| Native     | GSM8K    | 32    | 101.2% [99.6, 103.0]             |
| Native     | MATH     | 32    | 101.5% [99.4, 103.4]             |
| Retargeted | Code     | 32    | 102.1% [98.8, 105.6]             |
| Retargeted | Dialogue | 8     | 110.8% [107.8, 112.5]            |
| Retargeted | GSM8K    | 32    | 99.3% [96.8, 101.9]              |
| Retargeted | MATH     | 32    | 101.1% [96.7, 105.5]             |

We then repeat fitting at seed 1730 and compare both seeds on sixteen additional development conversations, disjoint from the initial eight. Within each worker, all four compact maps are paired. Mixed/math throughput ratios are 107.50% [104.34,110.44]% at seed 1729 and 107.52% [104.44,110.21]% at seed 1730 for retargeting. Native students give 101.09% [100.12,102.08]% and 100.28% [99.47,101.14]%, respectively. All 32 paired turn outputs match at each seed. Both seeds share the same conversations, so they are fitting replications rather than independent benchmark samples. This extension remains development evidence, with the same 256-token cap and no dialogue-quality scoring.

The fitting diagnostics help delimit the result. At seed 1729, mixed calibration lowers general-instruction validation error from 0.445 to 0.289 for retargeting and from 0.0468 to 0.0298 for native students. Both improve substantially in relative terms, but only retargeting shows a sizeable dialogue

throughput gain. Math errors increase from 0.227 to 0.251 and from 0.0241 to 0.0267, respectively. These diagnostics use the same 64 validation records per domain. The pattern suggests that domain coverage matters more for the harder retargeted interface in these tests. It also shows why proportional feature-error improvements alone do not predict proportional decoding gains.

## G FROZEN-CHECKPOINT GSM8K CONFIRMATION

We freeze the dense 512- and 2,048-example checkpoints at 8,192 updates and seed 1729 before generating confirmation outputs. The fixed test contains 256 GSM8K questions, sampled after exact and lexical exclusion against fitting data and collected evaluation history. A separate 256-question reserve was left unused by this experiment. Subsequently, 192 reserve questions were consumed by three separately frozen studies: 64 for the interface/SVD comparison and 64 different questions for native student confirmation, followed by 64 further questions for short-fit native confirmation (Appendix C). 64 reserve questions remain unused. Operational pilots use eight previously exposed GSM8K questions, not confirmation requests. The same confirmation questions are evaluated in both drafter families, so the second family is not an independent benchmark sample.

The primary speed criterion requires the lower endpoint of the paired 95% throughput-ratio interval to exceed 0.95. DFlash retains 97.48% [96.90,98.06]% and EAGLE-3 retains 98.90% [98.14,99.66]%. Throughput includes prompt processing. Resampling uses requests as paired units, 10,000 draws and seed 1729. These intervals condition on the frozen fitting checkpoints rather than estimating fitting-seed variation.

Table 57: All frozen-confirmation methods on 256 GSM8K requests per family. At cap counts outputs of exactly 2,048 tokens. Comparisons are within each family’s runtime.

| Family  | Method         | Tokens/s | Correct | At cap |
|---------|----------------|----------|---------|--------|
| DFlash  | AR             | 38.0     | 236/256 | 0      |
| DFlash  | Native drafter | 170.3    | 236/256 | 0      |
| DFlash  | Source reuse   | 99.0     | 236/256 | 0      |
| DFlash  | Dense 512      | 157.8    | 236/256 | 0      |
| DFlash  | Dense 2,048    | 161.9    | 236/256 | 0      |
| EAGLE-3 | AR             | 36.9     | 236/256 | 0      |
| EAGLE-3 | Native drafter | 100.9    | 238/256 | 0      |
| EAGLE-3 | Source reuse   | 70.7     | 238/256 | 0      |
| EAGLE-3 | Dense 512      | 100.0    | 238/256 | 0      |
| EAGLE-3 | Dense 2,048    | 101.1    | 238/256 | 0      |

The predeclared quality criterion requires the conservative paired 95% accuracy-difference lower bound relative to AR to exceed  $-1$  percentage point. DFlash has two beneficial and two adverse discordances, giving  $[-3.07, 3.07]$  points. EAGLE-3 has two beneficial and no adverse discordances, giving  $[-1.63, 3.14]$  points. Both quality outcomes are inconclusive under this criterion. Therefore the joint speed-and-quality criterion is not established, despite both speed criteria passing. Within each family, the two mapper sizes have identical per-question correctness, with a conservative interval of  $\pm 1.70$  points. DFlash also has identical token outputs between mapper sizes on all 256 requests, while EAGLE-3 has identical outputs on 254. Comparisons with AR retain the finite-precision distinctions discussed elsewhere. No reserve extension or outcome-dependent tuning is performed. Exclusion checks concern the recorded local history and lexical matches, not unknown pretraining exposure or semantic independence.

## H DATA CHECKS AND DEVELOPMENT EXPOSURE

The 4,096-record fitting manifest contains distinct problems with solutions. The loader renders both fields before applying the 192-token cap. Repeating the manifest changes presentations, not distinct records. The 32-question development set informed normalization and objective choices. The historical 468-question complement is computed from saved full-run outputs and retains that exposure history.

Normalized exact matching finds zero overlaps between the 4,096 fitting and 1,250 evaluation problem texts. For a second check, each text is split into unique normalized tokens. Token-set Jaccard overlap is the number of shared tokens divided by the number of tokens in either set. Each evaluation question is compared with its closest fitting record. Table 58 reports the resulting similarities.

Table 58: Closest fitting-to-evaluation token-set overlap. Threshold columns count evaluation records at or above the stated overlap.

| Task      | Maximum overlap | $\geq 0.80$ | $\geq 0.90$ | $\geq 0.95$ |
|-----------|-----------------|-------------|-------------|-------------|
| MATH-500  | 0.980           | 47          | 13          | 4           |
| GSM8K     | 0.378           | 0           | 0           | 0           |
| HumanEval | 0.290           | 0           | 0           | 0           |
| MBPP      | 0.333           | 0           | 0           | 0           |
| MT-Bench  | 0.571           | 0           | 0           | 0           |

MATH-500 includes 47 questions above 0.80 overlap and four above 0.95. These shared templates qualify the interpretation of math generalization. The fixed-map code and conversation evaluations give additional task coverage without changing the fitted map.

## I FITTING TIME AND ISOLATED MEMORY

Table 59: Selected map size and warm fitting-loop duration on four RTX 6000 Ada GPUs. Models are loaded before the recorded loop begins.

| Drafter | Target | Trainable weights (millions) | Fitting loop (seconds) |
|---------|--------|------------------------------|------------------------|
| DFlash  | 8B     | 52.43                        | 85.6                   |
| DFlash  | 14B    | 65.54                        | 118.0                  |
| EAGLE-3 | 8B     | 52.43                        | 86.7                   |
| EAGLE-3 | 14B    | 65.54                        | 118.5                  |

The fitting-loop timer covers source and target feature computation and matrix optimization. A complete setup also loads model weights, writes the map, reloads it and initializes generation. Published original training counts in Table 61 provide context for the additional-data requirement. They are not a wall-time or token-budget comparison with fitting a new native drafter.

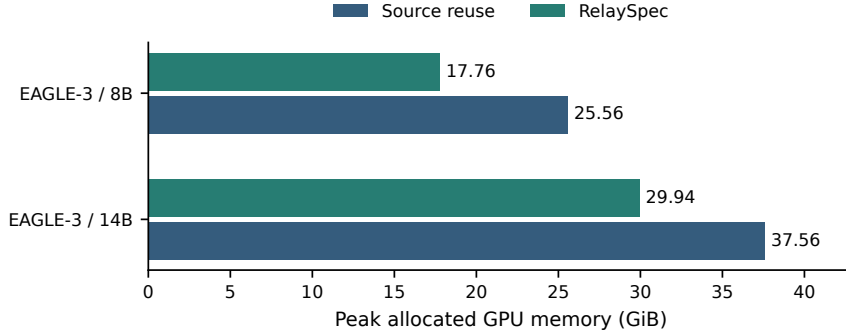


Figure 16: Isolated EAGLE-3 peak allocated memory on RTX 6000 Ada. Each method runs in a fresh process on eight paired prompts per target. Source removal saves 7.80 GiB at 8B and 7.63 GiB at 14B.

GPU peak counters are reset before the measured request. Allocated memory counts live tensor allocations and differs from reserved allocator memory. The mixed-arm processes used for timing are not the basis of this source-removal measurement.

## I.1 SELECTIVE FEATURE CAPTURE DURING GENERATION

A small conditioning interface also makes it unnecessary to retain all verifier-layer outputs. We implement selective forward hooks for the requested target taps, preserving the final normalization for the last layer. The transformer, mapper weights, verifier decisions and KV-cache policy are unchanged. This is an implementation consequence of the interface, not a new compression algorithm. Even the five-tap mapper benefits: reducing the tap count is not required for the basic capture optimization.

On Qwen3-8B/DFlash, we compare the same frozen checkpoint with full hidden-state return and selective capture. Each method has two repetitions in reversed order on one constructed archive prompt per length, with a 64-token output cap. Every pair matches output token hashes, output lengths, acceptance trajectories and target/draft calls. At 31,526 input tokens, the two-tap mapper reduces peak allocated memory from 32.73 to 25.03 GiB (23.5%), and the five-tap mapper from 32.98 to 25.87 GiB. A stricter control immediately releases prefill outputs and concatenated features after conditioning in both methods: savings remain 7.70 and 7.11 GiB, respectively. In that control, paired request time ratios range from 0.994 to 1.002 across the two lengths and maps, so these runs show memory savings without a material runtime change.

| Input tokens | Taps | Release prefill | All states (GiB) | Selected (GiB) | Saved (GiB) |
|--------------|------|-----------------|------------------|----------------|-------------|
| 2,273        | 5    | No              | 18.35            | 17.83          | 0.52        |
| 2,273        | 2    | No              | 18.22            | 17.66          | 0.56        |
| 8,993        | 5    | No              | 21.70            | 19.67          | 2.03        |
| 8,993        | 2    | No              | 21.54            | 19.35          | 2.20        |
| 16,826       | 5    | No              | 25.62            | 21.83          | 3.80        |
| 16,826       | 2    | No              | 25.43            | 21.33          | 4.11        |
| 31,526       | 5    | No              | 32.98            | 25.87          | 7.11        |
| 31,526       | 2    | No              | 32.73            | 25.03          | 7.70        |
| 16,826       | 5    | Yes             | 25.62            | 21.83          | 3.80        |
| 16,826       | 2    | Yes             | 25.43            | 21.32          | 4.11        |
| 31,526       | 5    | Yes             | 32.98            | 25.87          | 7.11        |
| 31,526       | 2    | Yes             | 32.73            | 25.03          | 7.70        |

Table 60: Selective-capture memory diagnostic on L40S. Values are mean peak allocated GiB over two order-reversed repetitions. These repeats are not independent prompts. Both frozen mapper copies are resident for both methods. Release-prefill rows control buffer lifetime after conditioning. Allocator-reserved memory is not the reported quantity.

A separate correctness check on eight code requests and eight two-turn dialogue conversations also preserves every output and acceptance trajectory for both the five-tap and two-tap mappers, with two order-reversed repetitions. These repeated executions do not add independent quality observations.

The synthetic inputs isolate sequence-length-dependent allocation and do not establish document understanding, long-context quality or concurrent-request capacity. Absolute values include model weights and the resident comparison maps. These results are distinct from the training-record scaling experiment and from the isolated source-removal experiment above.

## J ADDITIONAL RELATED WORK

**Improving drafting.** EAGLE predicts future features with shifted tokens (Li et al., 2024a). EAGLE-3 combines feature layers and simulated drafting steps (Li et al., 2025). DFlash predicts a block in parallel (Chen et al., 2026), while Medusa uses several prediction heads (Cai et al., 2024). GRIFFIN addresses training-to-decoding token alignment (Hu et al., 2025). RepSpec changes the training structure (Huo et al., 2026), while VSD and Draft-OPD optimize accepted proposals (Zou et al., 2026; Lei et al., 2026). RelaySpec inherits the trained drafter and changes only its conditioning input.

**Proposal structure and execution.** EAGLE-2 and OPT-Tree adapt proposal trees (Li et al., 2024b; Wang et al., 2025a). SpecInfer organizes tree verification (Miao et al., 2024). Distributed speculative inference and speculative speculative decoding overlap computation (Timor et al., 2025b; Kumar et al., 2026). Draft & Verify skips target layers (Zhang et al., 2024), LayerSkip trains early exits

(Elhoushi et al., 2024), and Lookahead checks candidates without an auxiliary drafter (Fu et al., 2024). RelaySpec retains an external drafter and holds its released proposal structure fixed. We evaluate the cost and accepted progress of changing its feature source.

## K INHERITED DRAFTER TRAINING CONTEXT

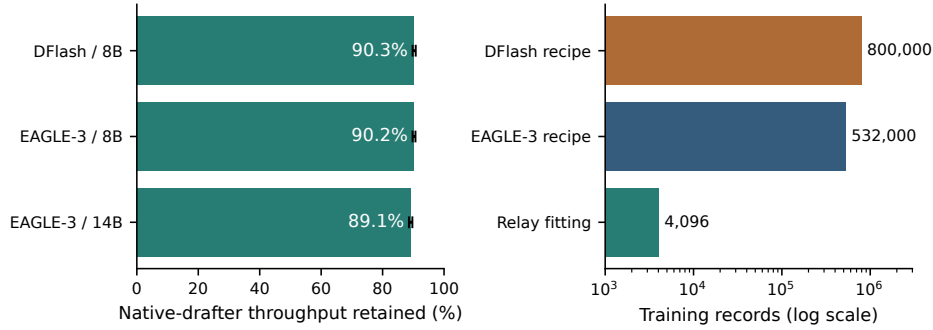


Figure 17: Left: fraction of native target-specific drafter throughput retained by RelaySpec in paired MATH-500 runs. Right: published original DFlash and EAGLE-3 recipe sizes versus the additional RelaySpec fitting set. Recipe counts provide training-scale context, not a matched training experiment or an audit of each released checkpoint.

Table 61: Original drafter training and additional target adaptation. Record counts describe different training stages and objectives.

| Stage                                       | Records  | Relative to relay fit | Weights learned |
|---------------------------------------------|----------|-----------------------|-----------------|
| Published DFlash recipe (Chen et al., 2026) | ~800,000 | ~195×                 | Drafter         |
| Published EAGLE-3 recipe (Li et al., 2025)  | ~532,000 | ~130×                 | Drafter         |
| RelaySpec target adaptation                 | 4,096    | 1×                    | Feature map     |

The DFlash recipe reports about 800K target-generated records. The EAGLE-3 recipe uses about 68K ShareGPT and 464K UltraChat entries. Relay fitting uses 0.51% and 0.77% of these respective record counts. This comparison describes marginal adaptation: the original drafter’s training remains inherited. Record lengths, objectives and hardware differ, so these ratios do not represent equivalent reductions in training compute. The evaluated EAGLE-3 checkpoints come from DeepSpec, while the table’s EAGLE-3 count refers specifically to the published training recipe.